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Connectivity Technologies – Part II

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6LoWPAN



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Introduction

- ✓ Low-power Wireless Personal Area Networks over IPv6.
- ✓ Allows for the smallest devices with limited processing ability to transmit information wirelessly using an Internet protocol.
- ✓ Allows low-power devices to connect to the Internet.
- ✓ Created by the Internet Engineering Task Force (IETF) - RFC 5933 and RFC 4919.

Source: T. Winter, P. Thubert, A. Brandt, J. Hui, R. Kelsey, P. Levis, K. Pister, R. Struik, JP. Vasseur, R. Alexander, "[RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks](#)", IETF, Standards Track, Mar. 2012



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Features of 6LoWPANs

- ✓ Allows IEEE 802.15.4 radios to carry 128-bit addresses of Internet Protocol version 6 (IPv6).
- ✓ Header compression and address translation techniques allow the IEEE 802.15.4 radios to access the Internet.
- ✓ IPv6 packets compressed and reformatted to fit the IEEE 802.15.4 packet format.
- ✓ Uses include IoT, Smart grid, and M2M applications.



Addressing in 6LoWPAN

Addressing

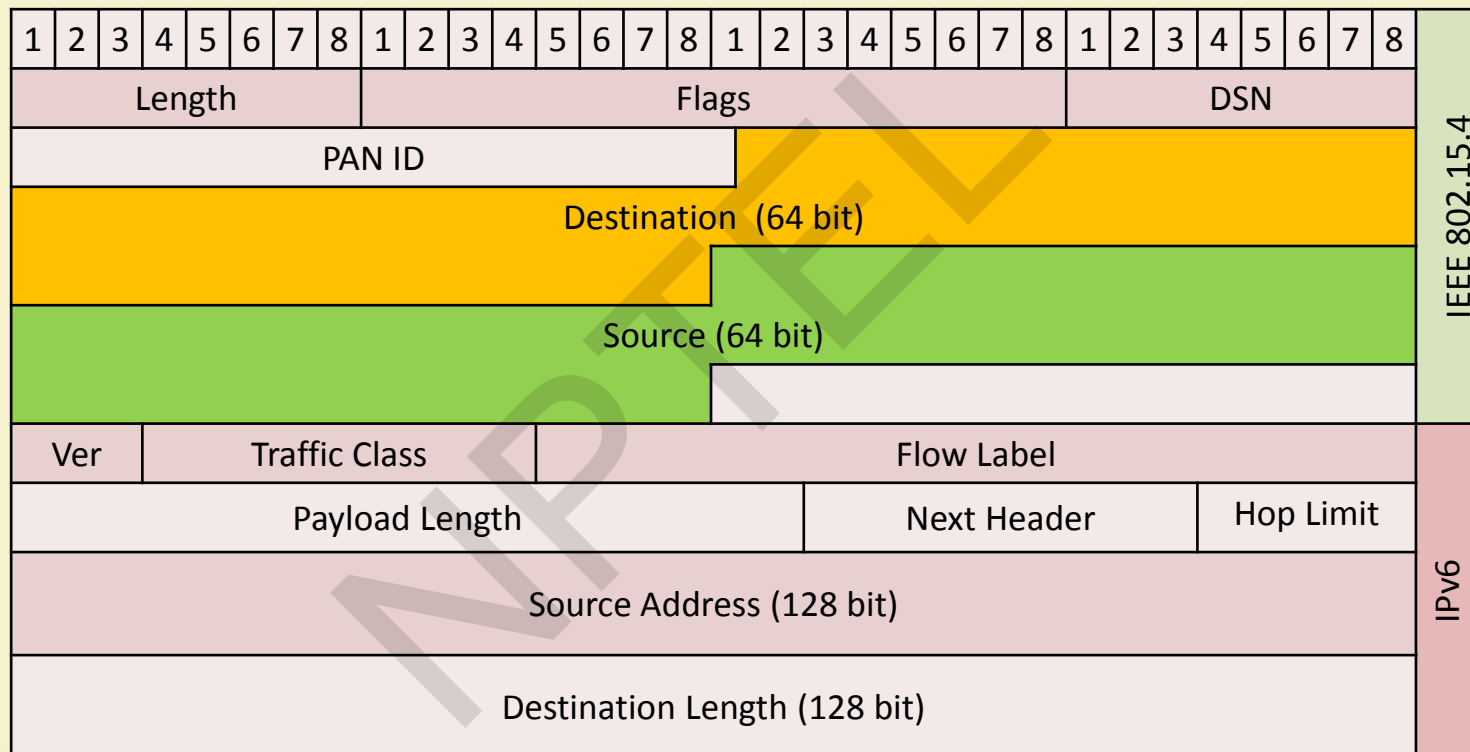
64-bit
Extended

16-bit
Short

- 64-bit addresses: globally unique
- 16 bit addresses: PAN specific; assigned by PAN coordinator
- IPv6 multicast not supported by 802.15.4
- IPv6 packets carried as link layer broadcast frames



6LoWPAN Packet Format



Header Type: Dispatch Header

1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
0	1	Dispatch						Type Specific Header																							

- **Dispatch:** Initiates communication
- **0,1:** Identifier for Dispatch Type
- **Dispatch:**
 - 6 bits
 - Identifies the next header type
- **Type Specific Header:**
 - Determined by Dispatch header



Header Type: Mesh Addressing Header

1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
1 0		V	F	Hops Left				Originator Address												Final Address											

- **1,0:** ID for Mesh Addressing Header
- **V:** '0' if originator is 64-bit extended address, '1' if 16-bit address
- **F:** '0' if destination is 64-bit addr., '1' if 16-bit addr.
- **Hops Left:** decremented by each node before sending to next hop



Header Type: Fragmentation Header

1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
1 1 0 0				Datagram Size														Datagram Tag													

(a) First Fragment

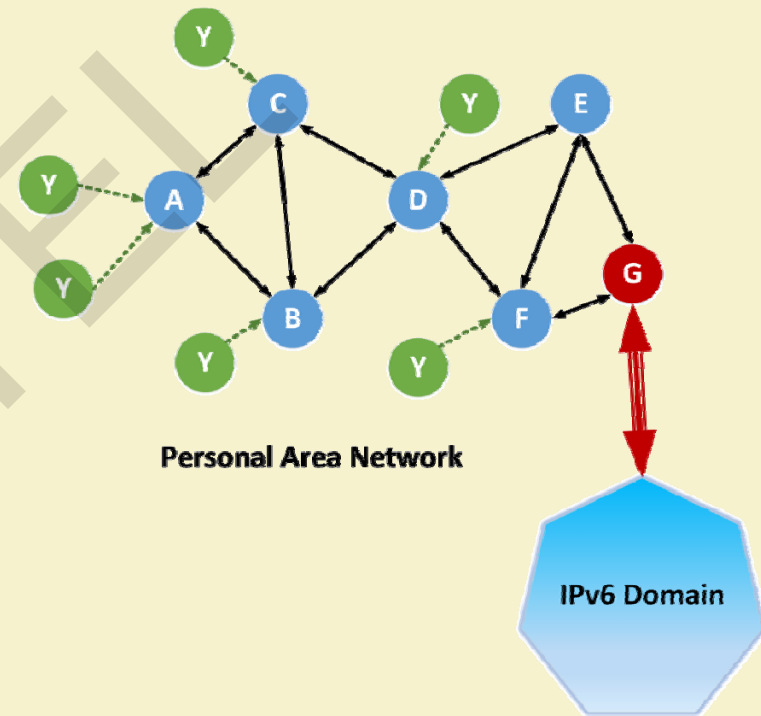
1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8			
1 1 0 0				Datagram Size														Datagram Tag																
Datagram Offset																																		

(b) Subsequent Fragment



6LoWPAN Routing Considerations

- ✓ Mesh routing within the PAN space.
- ✓ Routing between IPv6 and the PAN domain
- ✓ Routing protocols in use:
 - LOADng
 - RPL



LOADng Routing

- ✓ Derived from AODV and extended for use in IoT.
- ✓ Basic operations of LOADng include:
 - Generation of **Route Requests (RREQs)** by a LOADng Router (originator) for discovering a route to a destination,
 - **Forwarding of such RREQs** until they reach the destination LOADng Router,
 - Generation of **Route Replies (RREPs)** upon receipt of an RREQ by the indicated destination, and unicast hop-by-hop forwarding of these RREPs towards the originator.

Source: Clausen, T.; Colin de Verdiere, A.; Yi, J.; Niktash, A.; Igarashi, Y.; Satoh, H.; Herberg, U.; Lavenu, C. et al. (January 2016). [The Lightweight On-demand Ad hoc Distance-vector Routing Protocol - Next Generation \(LOADng\)](#). IETF. I-D draft-clausen-ltn-loadng-14



- If a route is detected to be broken, a **Route Error (RERR)** message is returned to the originator of that data packet to inform the originator about the route breakage.
- **Optimized flooding** is supported, reducing the overhead incurred by RREQ generation and flooding.
- Only the destination is permitted to respond to an RREQ.
- Intermediate LOADng Routers are explicitly prohibited from responding to RREQs, even if they may have active routes to the sought destination.
- RREQ/RREP messages generated by a given LOADng Router share a single unique, monotonically increasing sequence number.

Source: Clausen, T.; Colin de Verdiere, A.; Yi, J.; Niktash, A.; Igarashi, Y.; Satoh, H.; Herberg, U.; Lavenu, C. et al. (January 2016). [The Lightweight On-demand Ad hoc Distance-vector Routing Protocol - Next Generation \(LOADng\)](#). [IETF](#). I-D draft-clausen-lln-loadng-14



RPL Routing

- ✓ Distance Vector IPv6 **routing protocol for lossy and low power networks.**
- ✓ Maintains routing topology using low rate beaconing.
- ✓ Beaconing rate increases on detecting inconsistencies (e.g. node/link in a route is down).
- ✓ Routing information included in the datagram itself.
- ✓ **Proactive:** Maintaining routing topology.
- ✓ **Reactive:** Resolving routing inconsistencies.

Source: T. Winter, P. Thubert, A. Brandt, J. Hui, R. Kelsey, P. Levis, K. Pister, R. Struik, JP. Vasseur, R. Alexander, "[RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks](#)", IETF, Standards Track, Mar. 2012



- ✓ RPL separates packet processing and forwarding from the routing optimization objective, which helps in Low power Lossy Networks (LLN).
- ✓ RPL supports message confidentiality and integrity.
- ✓ Supports Data-Path Validation and Loop Detection
- ✓ Routing optimization objectives include
 - minimizing energy
 - minimizing latency
 - satisfying constraints (w.r.t node power, bandwidth, etc.)

Source: T. Winter, P. Thubert, A. Brandt, J. Hui, R. Kelsey, P. Levis, K. Pister, R. Struik, JP. Vasseur, R. Alexander, "[RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks](#)", IETF, Standards Track, Mar. 2012



- ✓ RPL operations require bidirectional links.
- ✓ In some LLN scenarios, those links may exhibit asymmetric properties.
- ✓ It is required that the reachability of a router be verified before the router can be used as a parent.

Source: T. Winter, P. Thubert, A. Brandt, J. Hui, R. Kelsey, P. Levis, K. Pister, R. Struik, JP. Vasseur, R. Alexander, "[RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks](#)", IETF, Standards Track, Mar. 2012



RFID



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- ✓ RFID is an acronym for “radio-frequency identification”
- ✓ Data digitally encoded in RFID tags, which can be read by a reader.
- ✓ Somewhat similar to barcodes.
- ✓ Data read from tags are stored in a database by the reader.
- ✓ As compared to traditional barcodes and QR codes, RFID tag data can be read outside the line-of-sight.

Source: “[How does RFID work?](#)” AB&R (Online)



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RFID Features

- ✓ RFID tag consists of an integrated circuit and an antenna.
- ✓ The tag is covered by a protective material which also acts as a shield against various environmental effects.
- ✓ Tags may be passive or active.
- ✓ Passive RFID tags are the most widely used.
- ✓ Passive tags have to be powered by a reader inductively before they can transmit information, whereas active tags have their own power supply.

Source: "[How does RFID work?](#)" AB&R (Online)



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Working Principle

- ✓ Derived from Automatic Identification and Data Capture (AIDC) technology.
- ✓ AIDC performs object identification, object data collection and mapping of the collected data to computer systems with little or no human intervention.
- ✓ AIDC uses wired communication
- ✓ RFID uses radio waves to perform AIDC functions.
- ✓ The main components of an RFID system include an RFID tag or smart label, an RFID reader, and an antenna.

Source: "[How does RFID work?](#)" AB&R (Online)

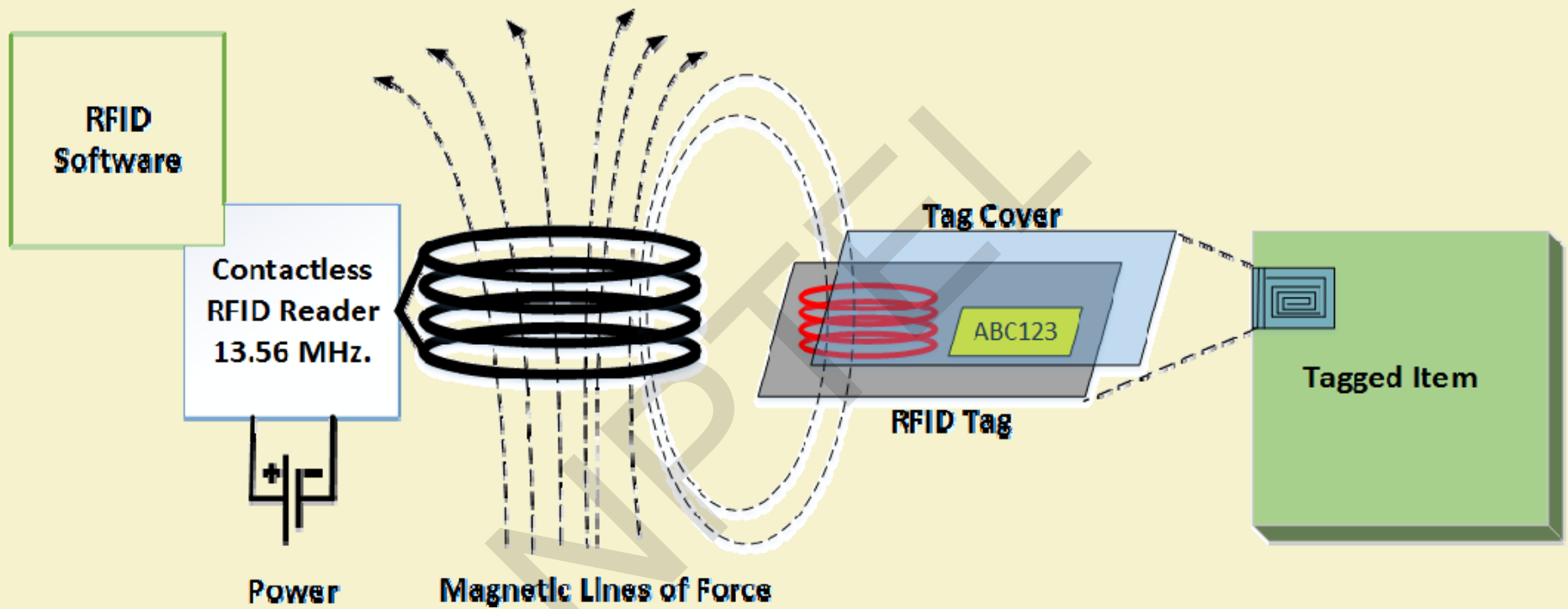


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Applications

- ✓ Inventory management
- ✓ Asset tracking
- ✓ Personnel tracking
- ✓ Controlling access to restricted areas
- ✓ ID badging
- ✓ Supply chain management
- ✓ Counterfeit prevention (e.g. in the pharmaceutical industry)

Source: "[How does RFID work?](#)" AB&R (Online)



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Basics of IoT Networking – Part II

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Functionality-based IoT Protocol Organization

- ✓ **Connectivity** (6LowPAN, RPL)
- ✓ **Identification** (EPC, uCode, IPv6, URIs)
- ✓ **Communication / Transport** (WiFi, Bluetooth, LPWAN)
- ✓ **Discovery** (Physical Web, mDNS, DNS-SD)
- ✓ **Data Protocols** (MQTT, CoAP, AMQP, Websocket, Node)
- ✓ **Device Management** (TR-069, OMA-DM)
- ✓ **Semantic** (JSON-LD, Web Thing Model)
- ✓ **Multi-layer Frameworks** (Alljoyn, IoTivity, Weave, Homekit)

Source: [Internet of Things Protocols \(Online\)](#)



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MQTT



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Introduction

- ✓ **Message Queue Telemetry Transport.**
- ✓ ISO standard (ISO/IEC PRF 20922).
- ✓ It is a publish-subscribe-based lightweight messaging protocol for use in conjunction with the TCP/IP protocol.
- ✓ MQTT was introduced by IBM in 1999 and standardized by OASIS in 2013.
- ✓ Designed to provide connectivity (mostly embedded) between applications and middle-wares on one side and networks and communications on the other side.

Source: [“MQTT”, Wikipedia \(Online\)](#)



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- ✓ A message broker controls the publish-subscribe messaging pattern.
- ✓ A topic to which a client is subscribed is updated in the form of messages and distributed by the message broker.
- ✓ Designed for:
 - Remote connections
 - Limited bandwidth
 - Small-code footprint

Source: [“MQTT”, Wikipedia \(Online\)](#)



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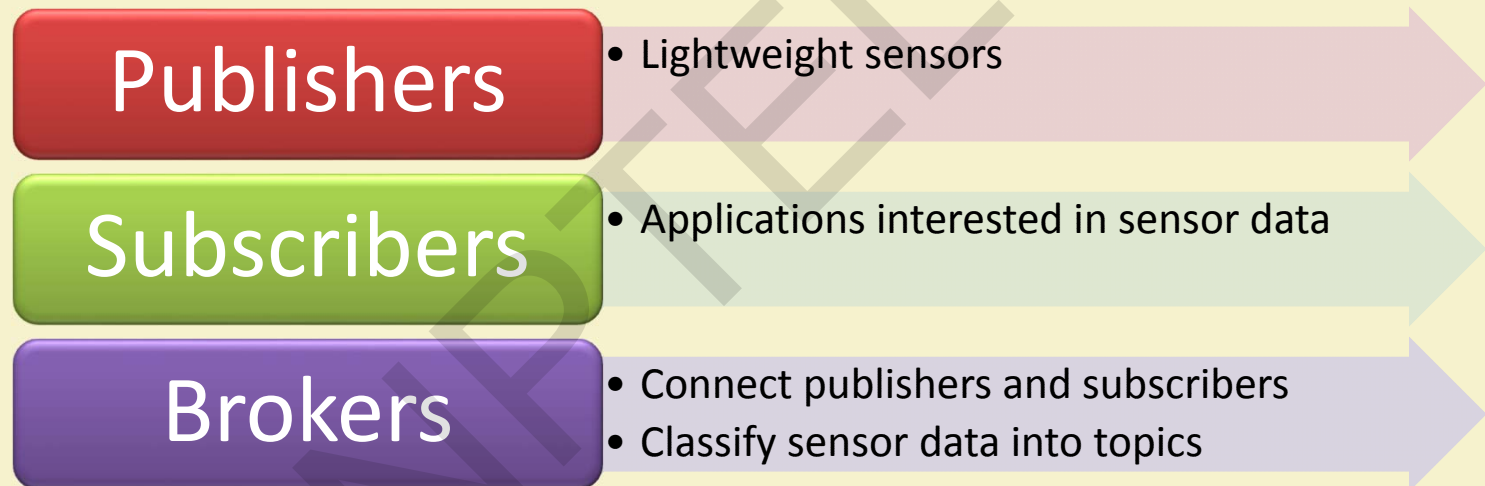


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MQTT Components



Source: [“MQTT”, Wikipedia \(Online\)](#)



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MQTT Methods

Connect

Disconnect

Subscribe

Unsubscribe

Publish

Source: [“MQTT”, Wikipedia \(Online\)](#)



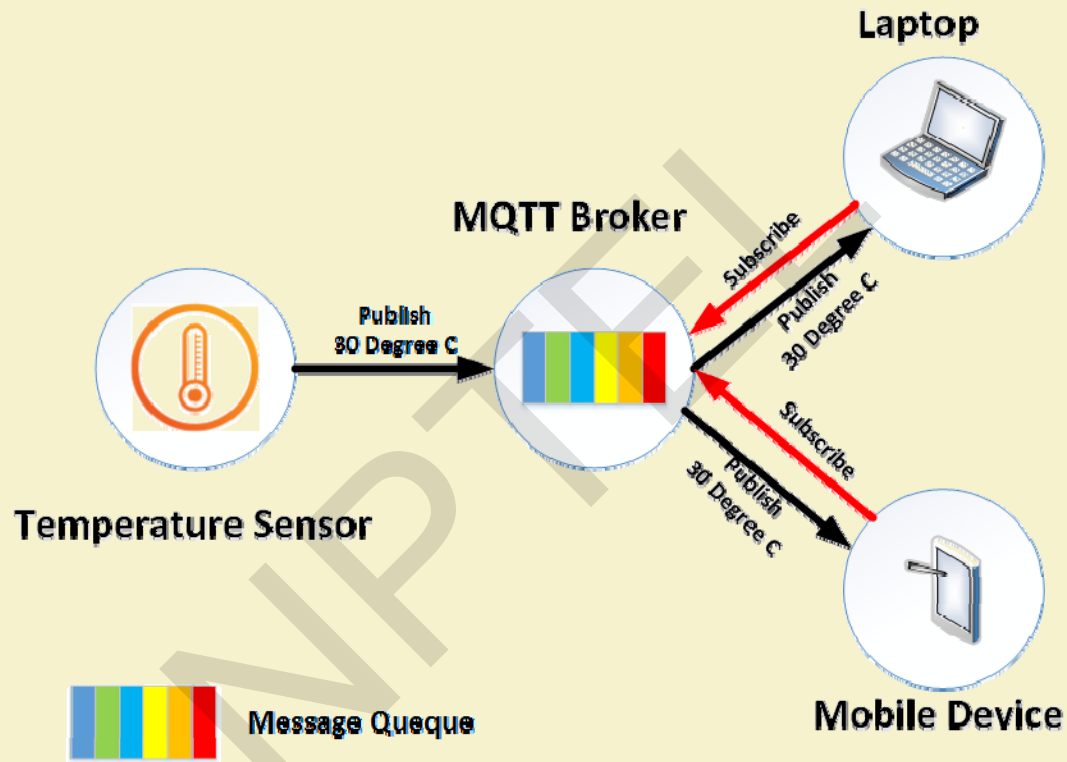
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Source: [“MQTT 101 – How to Get Started with the lightweight IoT Protocol”, HiveMQ \(Online\)](#)



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Communication

- ✓ The protocol uses a **publish/subscribe** architecture (HTTP uses a request/response paradigm).
- ✓ Publish/subscribe is **event-driven** and enables messages to be pushed to clients.
- ✓ The central **communication point is the MQTT broker**, which is in charge of dispatching all messages between the senders and the rightful receivers.
- ✓ Each client that publishes a message to the broker, includes a **topic** into the message. The **topic is the routing information for the broker**.

Source: [“MQTT 101 – How to Get Started with the lightweight IoT Protocol”, HiveMQ \(Online\)](#)



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- ✓ Each client that wants to receive messages subscribes to a certain topic and the broker delivers all messages with the matching topic to the client.
- ✓ Therefore the clients don't have to know each other. They only communicate over the topic.
- ✓ This architecture enables highly scalable solutions without dependencies between the data producers and the data consumers.

Source: [“MQTT 101 – How to Get Started with the lightweight IoT Protocol”, HiveMQ \(Online\)](#)



MQTT Topics

- ✓ A topic is a **simple string** that can have more hierarchy levels, which are separated by a slash.
- ✓ A sample topic for sending temperature data of the living room could be *house/living-room/temperature*.
- ✓ On one hand the client (e.g. mobile device) can subscribe to the exact topic or on the other hand, it can use a **wildcard**.

Source: [“MQTT 101 – How to Get Started with the lightweight IoT Protocol”, HiveMQ \(Online\)](#)



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- ✓ The subscription to *house/+/temperature* would result in all messages sent to the previously mentioned topic *house/living-room/temperature*, as well as any topic with an arbitrary value in the place of living room, such as *house/kitchen/temperature*.
- ✓ The plus sign is a **single level wild card** and only allows arbitrary values for one hierarchy.
- ✓ If more than one level needs to be subscribed, such as, the entire sub-tree, there is also a **multilevel wildcard (#)**.
- ✓ It allows to subscribe to all underlying hierarchy levels.
- ✓ For example *house/#* is subscribing to all topics beginning with *house*.

Source: [“MQTT 101 – How to Get Started with the lightweight IoT Protocol”, HiveMQ \(Online\)](#)



Applications

- ✓ **Facebook Messenger** uses MQTT for online chat.
- ✓ **Amazon Web Services** use Amazon IoT with MQTT.
- ✓ **Microsoft Azure** IoT Hub uses MQTT as its main protocol for telemetry messages.
- ✓ The **EVERYTHING IoT platform** uses MQTT as an M2M protocol for millions of connected products.
- ✓ **Adafruit** launched a free MQTT cloud service for IoT experimenters called Adafruit IO.



SMQTT

- ✓ **Secure MQTT** is an extension of MQTT which uses encryption based on lightweight attribute based encryption.
- ✓ The main advantage of using such encryption is the broadcast encryption feature, in which one message is encrypted and delivered to multiple other nodes, which is quite common in IoT applications.
- ✓ In general, the algorithm consists of four main stages: setup, encryption, publish and decryption.

Source: M. Singh, M. Rajan, V. Shivraj, and P. Balamuralidhar, "Secure MQTT for Internet of Things (IoT)," in Fifth International Conference on Communication Systems and Network Technologies (CSNT 2015), April 2015, pp. 746-751



- ✓ In the setup phase, the subscribers and publishers register themselves to the broker and get a master secret key according to their developer's choice of key generation algorithm.
- ✓ When the data is published, it is encrypted and published by the broker which sends it to the subscribers, which is finally decrypted at the subscriber end having the same master secret key.
- ✓ The key generation and encryption algorithms are not standardized.
- ✓ SMQTT is proposed only to enhance MQTT security features.

Source: M. Singh, M. Rajan, V. Shivraj, and P. Balamuralidhar, "Secure MQTT for Internet of Things (IoT)," in Fifth International Conference on Communication Systems and Network Technologies (CSNT 2015), April 2015, pp. 746-751



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Basics of IoT Networking – Part III

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CoAP



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- ✓ CoAP – **Constrained Application Protocol**.
- ✓ **Web transfer protocol** for use with constrained nodes and networks.
- ✓ **Designed for Machine to Machine (M2M)** applications such as smart energy and building automation.
- ✓ Based on **Request-Response model** between end-points
- ✓ Client-Server interaction is **asynchronous over a datagram oriented transport protocol** such as UDP

Source: Z. Shelby , K. Hartke, C. Bormann, “**The Constrained Application Protocol (CoAP)**”, Internet Engineering Task Force (IETF), Standards Track, 2014



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- ✓ The Constrained Application Protocol (CoAP) is a session layer protocol designed by IETF Constrained RESTful Environment (CoRE) working group to provide lightweight RESTful (HTTP) interface.
- ✓ Representational State Transfer (REST) is the standard interface between HTTP client and servers.
- ✓ Lightweight applications such as those in IoT, could result in significant overhead and power consumption by REST.
- ✓ CoAP is designed to enable low-power sensors to use RESTful services while meeting their power constraints.

Source: Z. Shelby , K. Hartke, C. Bormann, “**The Constrained Application Protocol (CoAP)**”, Internet Engineering Task Force (IETF), Standards Track, 2014

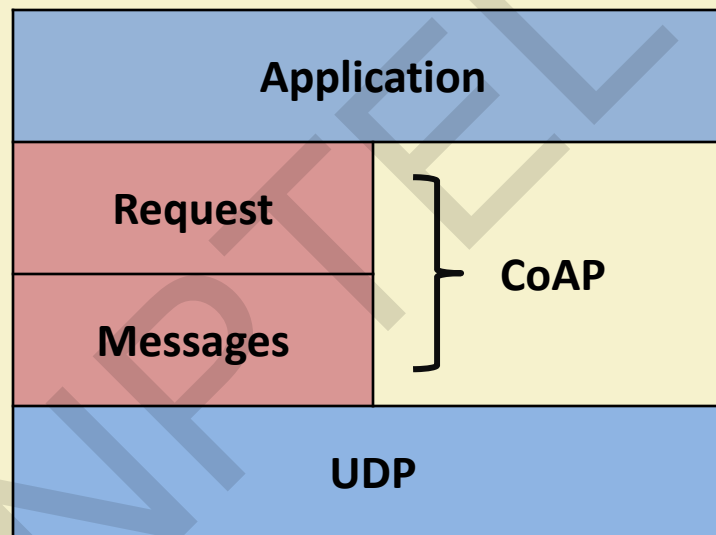


- ✓ Built over UDP, instead of TCP (which is commonly used with HTTP) and has a light mechanism to provide reliability.
- ✓ CoAP architecture is divided into two main sub-layers:
 - Messaging
 - Request/response.
- ✓ The messaging sub-layer is responsible for reliability and duplication of messages, while the request/response sub-layer is responsible for communication.
- ✓ CoAP has four messaging modes:
 - Confirmable
 - Non-confirmable
 - Piggyback
 - Separate

Source: V. Karagiannis, P. Chatzimisios, F. Vazquez-Gallego, and J. Alonso-Zarate, "A survey on application layer protocols for the internet of things," Transaction on IoT and Cloud Computing, vol. 3, no. 1, pp. 11-17, 2015



CoAP Position



Source: Z. Shelby , K. Hartke, C. Bormann, **“The Constrained Application Protocol (CoAP)”**, Internet Engineering Task Force (IETF), Standards Track, 2014



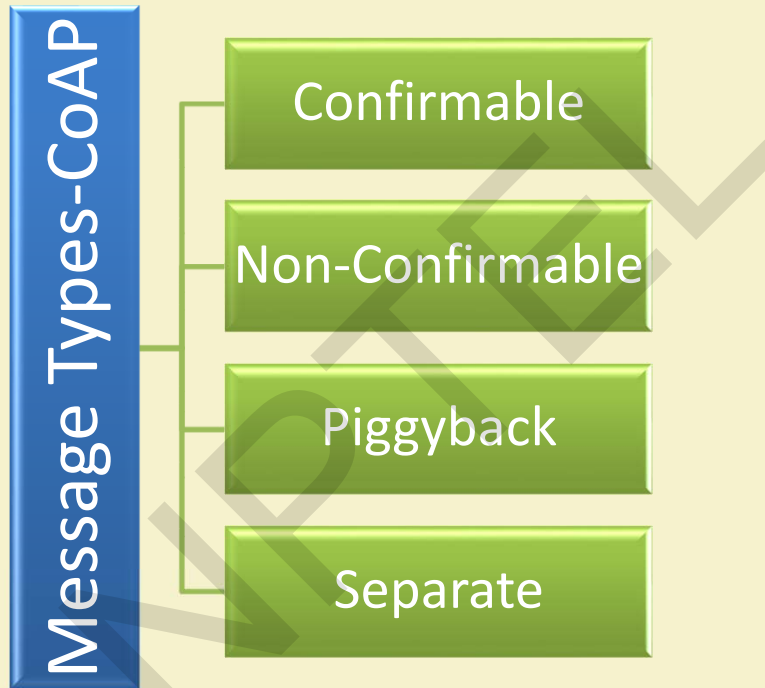
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CoAP Message Types



Source: Z. Shelby , K. Hartke, C. Bormann, **"The Constrained Application Protocol (CoAP)"**, Internet Engineering Task Force (IETF), Standards Track, 2014



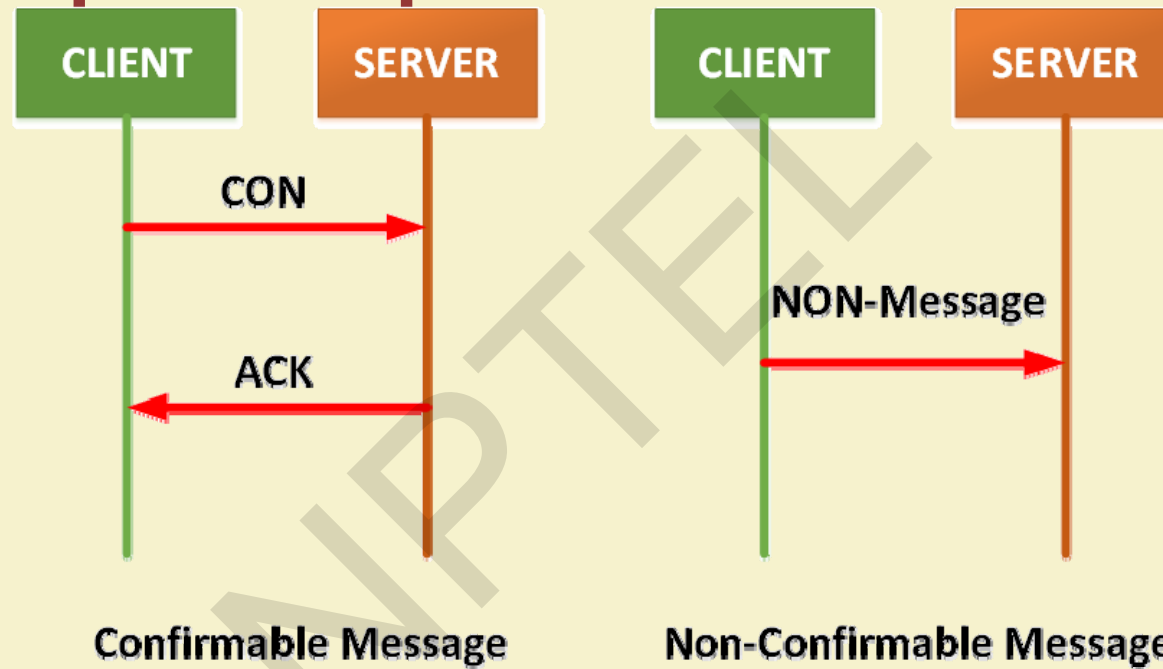
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CoAP Request-Response Model



Source: V. Karagiannis, P. Chatzimisios, F. Vazquez-Gallego, and J. Alonso-Zarate, "A survey on application layer protocols for the internet of things," Transaction on IoT and Cloud Computing, vol. 3, no. 1, pp. 11-17, 2015



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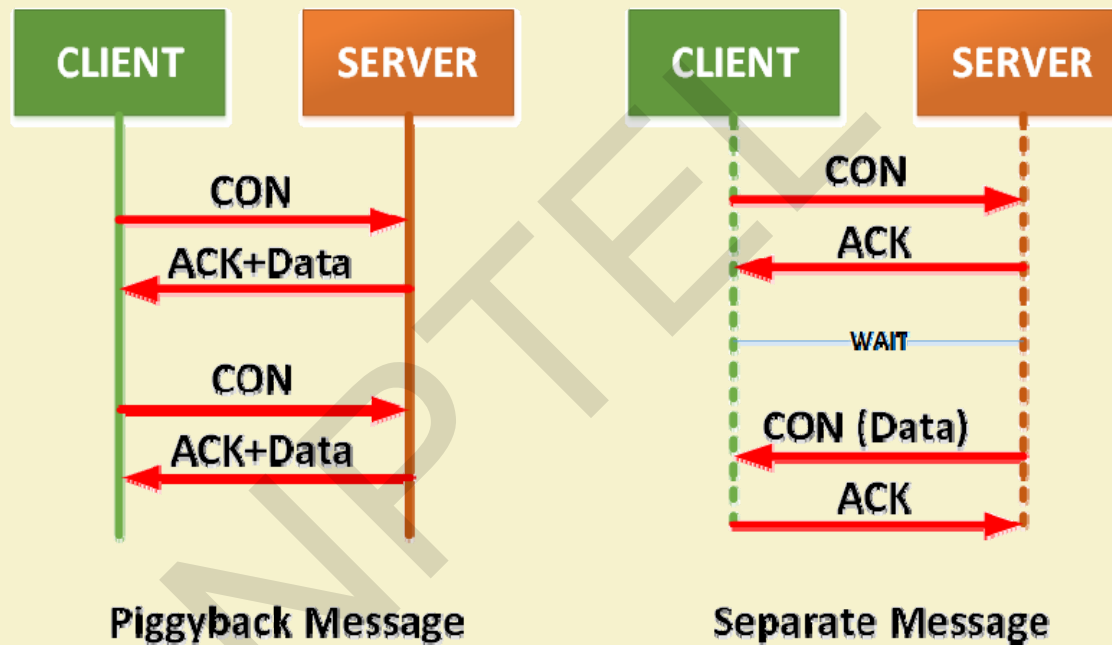
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- ✓ Confirmable and non-confirmable modes represent the reliable and unreliable transmissions, respectively, while the other modes are used for request/response.
- ✓ Piggyback is used for client/server direct communication where the server sends its response directly after receiving the message, i.e., within the acknowledgment message.
- ✓ On the other hand, the separate mode is used when the server response comes in a message separate from the acknowledgment, and may take some time to be sent by the server.
- ✓ Similar to HTTP, CoAP utilizes GET, PUT, PUSH, DELETE messages requests to retrieve, create, update, and delete, respectively

Source: V. Karagiannis, P. Chatzimisios, F. Vazquez-Gallego, and J. Alonso-Zarate, "A survey on application layer protocols for the internet of things," Transaction on IoT and Cloud Computing, vol. 3, no. 1, pp. 11-17, 2015



CoAP Request-Response Model



Source: V. Karagiannis, P. Chatzimisios, F. Vazquez-Gallego, and J. Alonso-Zarate, "A survey on application layer protocols for the internet of things," Transaction on IoT and Cloud Computing, vol. 3, no. 1, pp. 11-17, 2015



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Features

- ✓ Reduced overheads and parsing complexity.
- ✓ URL and content-type support.
- ✓ Support for the discovery of resources provided by known CoAP services.
- ✓ Simple subscription for a resource, and resulting push notifications.
- ✓ Simple caching based on maximum message age.

Source: ["Constrained Application Protocol", Wikipedia \(Online\)](#)



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XMPP



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Introduction

- ✓ **XMPP – Extensible Messaging and Presence Protocol.**
- ✓ A communication protocol for **message-oriented middleware** based on XML (Extensible Markup Language).
- ✓ Real-time exchange of structured data.
- ✓ It is an open standard protocol.

Source: [“XMPP”, Wikipedia \(Online\)](#)



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- ✓ XMPP uses a **client-server architecture**.
- ✓ As the model is **decentralized**, no central server is required.
- ✓ XMPP provides for the **discovery of services** residing locally or across a network, and the **availability information** of these services.
- ✓ Well-suited for cloud computing where virtual machines, networks, and firewalls would otherwise present obstacles to alternative service discovery and presence-based solutions.
- ✓ Open means to support machine-to-machine or peer-to-peer communications across a diverse set of networks.

Source: [“XMPP”, Wikipedia \(Online\)](#)



Highlights

- ✓ Decentralization – No central server; anyone can run their own XMPP server.
- ✓ Open standards – No royalties or granted permissions are required to implement these specifications
- ✓ Security – Authentication, encryption, etc.
- ✓ Flexibility – Supports interoperability

Source: [“XMPP”, Wikipedia \(Online\)](#)

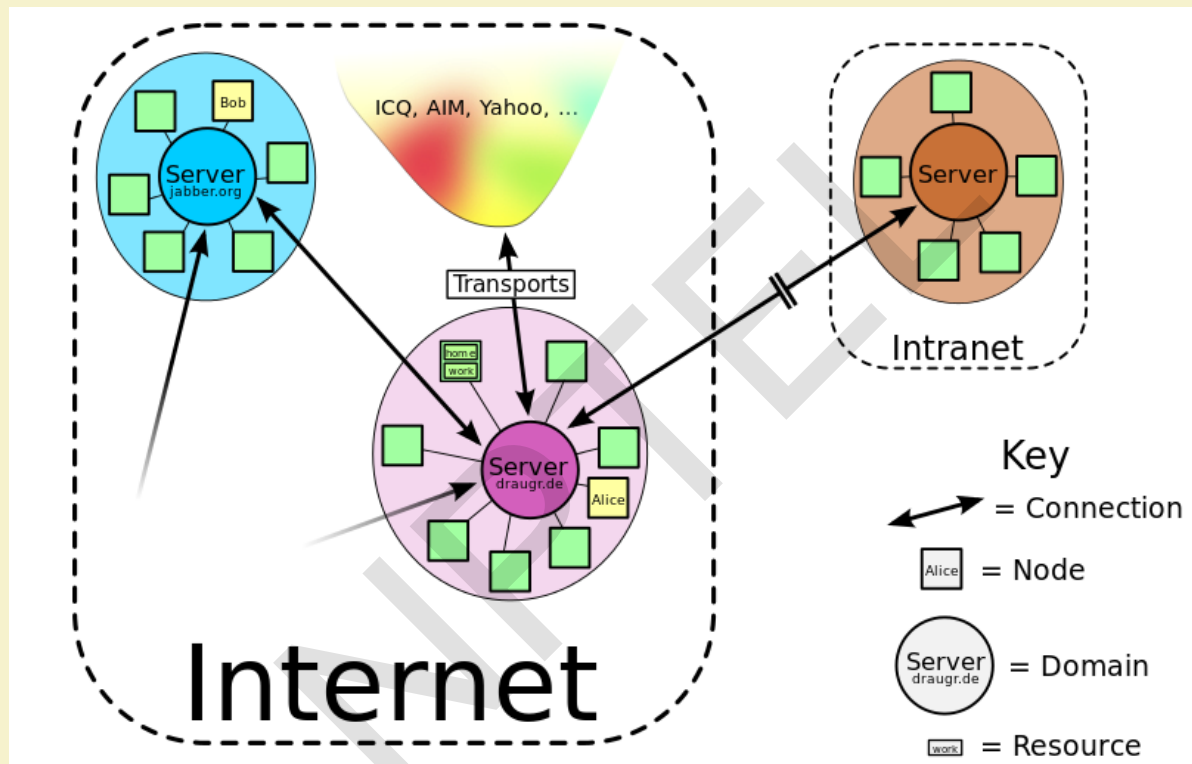


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Source: ["JabberNetwork.svg"](http://JabberNetwork.svg), Wikimedia Commons (Online)



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Core XMPP Technologies

Core

- information about the core XMPP technologies for XML streaming

Jingle

- multimedia signalling for voice, video, file transfer

Multi-user Chat

- flexible, multi-party communication

PubSub

- alerts and notifications for data syndication

BOSH

- HTTP binding for XMPP

Source: ["XMPP: Technology Overview", XMPP.org \(Online\)](http://xmpp.org)



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Weaknesses

- ✓ Does not support QoS.
- ✓ Text based communications induces higher network overheads.
- ✓ Binary data must be first encoded to **base64** before transmission.



Applications

- ✓ Publish-subscribe systems
- ✓ Signaling for VoIP
- ✓ Video
- ✓ File transfer
- ✓ Gaming
- ✓ Internet of Things applications
 - Smart grid
 - Social networking services



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Basics of IoT Networking – Part IV

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AMQP



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Introduction

- ✓ **Advanced Message Queuing Protocol.**
- ✓ **Open standard for passing business messages** between applications or organizations.
- ✓ Connects between systems and business processes.
- ✓ It is a binary application layer protocol.
- ✓ Basic unit of data is a *frame*.
- ✓ ISO standard: **ISO/IEC 19464**

Source: [“Advanced Message Queuing Protocol”, Wikipedia \(Online\)](#)



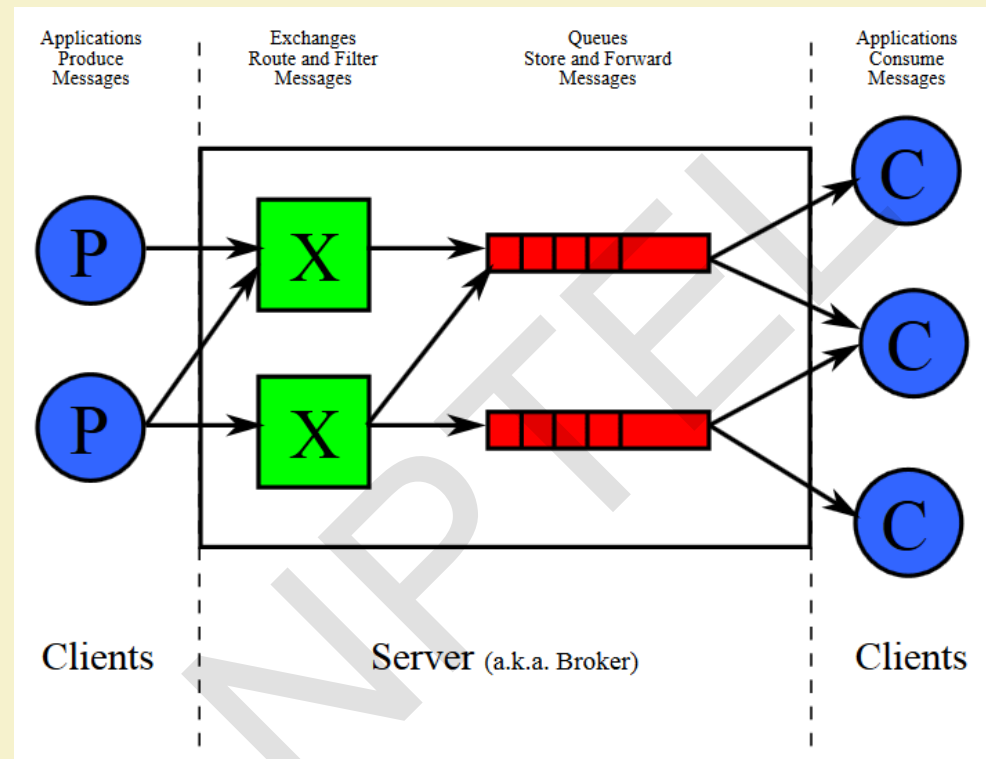
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Source: ["The-amqp-model-for-wikipedia.svg", Wikimedia Commons \(Online\)](#)



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4

AMQP Features



Features

Security

Reliability

Interoperability

Routing

Queuing

Open standard



Message Delivery Guarantees

- ✓ *At-most-once*
 - each message is delivered once or never
- ✓ *At-least-once*
 - each message is certain to be delivered, but may do so multiple times
- ✓ *Exactly-once*
 - message will always certainly arrive and do so only once

Reference: "OASIS AMQP version 1.0, sections 2.6.12-2.6.13". OASIS AMQP Technical Committee



AMQP Frame Types

- ✓ Nine AMQP frame types are defined that are used to initiate, control and tear down the transfer of messages between two peers:
 - Open (connection open)
 - Begin (session open)
 - Attach (initiate new link)
 - Transfer (for sending actual messages)
 - Flow (controls message flow rate)
 - Disposition (Informs the changes in state of transfer)
 - Detach (terminate the link)
 - End (session close)
 - Close (connection close)

Source: [O.S. Tezer, "An advanced messaging queuing protocol walkthrough", DigitalOcean \(Online\), 2013](#)



Components

Exchange

- Part of Broker
- Receives messages and routes them to Queues

Queue

- Separate queues for separate business processes
- Consumers receive messages from queues

Bindings

- Rules for distributing messages (who can access what message, destination of the message)

Source: [O.S. Tezer, "An advanced messaging queuing protocol walkthrough", DigitalOcean \(Online\), 2013](#)



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AMQP Exchanges

Direct

Fan-out

Topic

Header

Source: [O.S. Tezer, "An advanced messaging queuing protocol walkthrough", DigitalOcean \(Online\), 2013](#)



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AMQP Features

- ✓ Targeted QoS (Selectively offering QoS to links)
- ✓ Persistence (Message delivery guarantees)
- ✓ Delivery of messages to multiple consumers
- ✓ Possibility of ensuring multiple consumption
- ✓ Possibility of preventing multiple consumption
- ✓ High speed protocol

Source: [O.S. Tezer, "An advanced messaging queuing protocol walkthrough", DigitalOcean \(Online\), 2013](#)



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Applications

- ✓ Monitoring and global update sharing.
- ✓ Connecting different systems and processes to talk to each other.
- ✓ Allowing servers to respond to immediate requests quickly and delegate time consuming tasks for later processing.
- ✓ Distributing a message to multiple recipients for consumption.
- ✓ Enabling offline clients to fetch data at a later time.
- ✓ Introducing fully asynchronous functionality for systems.
- ✓ Increasing reliability and uptime of application deployments.

Source: [O.S. Tezer, "An advanced messaging queuing protocol walkthrough", DigitalOcean \(Online\), 2013](#)



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Connectivity Technologies – Part I

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Communication Protocols

The following communication protocols have immediate importance to consumer and industrial IoTs:

- ✓ IEEE 802.15.4
- ✓ Zigbee
- ✓ 6LoWPAN
- ✓ Wireless HART
- ✓ Z-Wave
- ✓ ISA 100
- ✓ Bluetooth
- ✓ NFC
- ✓ RFID



IEEE 802.15.4



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Features of IEEE 802.15.4

- ✓ Well-known standard for low data-rate WPAN.
- ✓ Developed for low-data-rate monitoring and control applications and extended-life low-power-consumption uses.
- ✓ This standard uses only the first two layers (PHY, MAC) plus the logical link control (LLC) and service specific convergence sub-layer (SSCS) additions to communicate with all upper layers
- ✓ Operates in the ISM band.

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



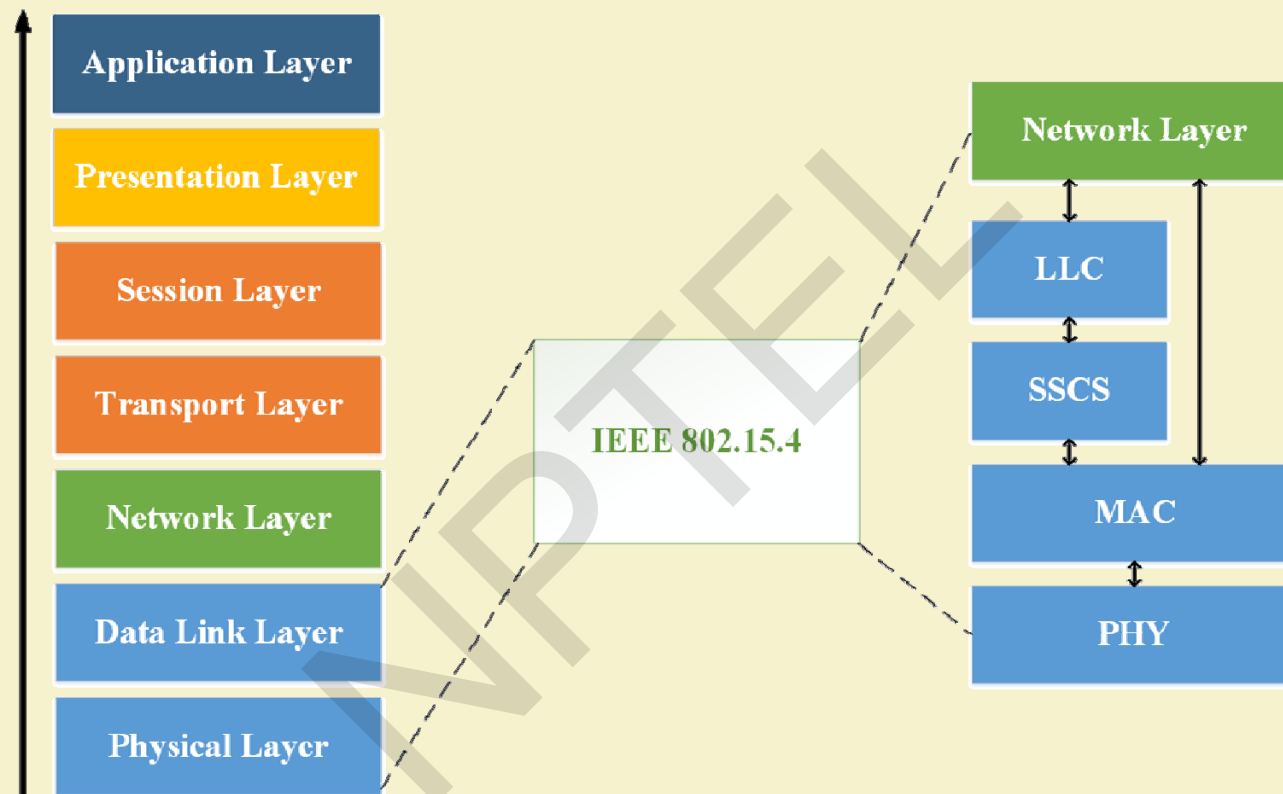
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- ✓ Uses direct sequence spread spectrum (DSSS) modulation.
- ✓ Highly tolerant of noise and interference and offers link reliability improvement mechanisms.
- ✓ Low-speed versions use Binary Phase Shift Keying (BPSK).
- ✓ High data-rate versions use offset-quadrature phase-shift keying (O-QPSK).
- ✓ Uses carrier sense multiple access with collision avoidance (CSMA-CA) for channel access.
- ✓ Multiplexing allows multiple users or nodes interference-free access to the same channel at different times.

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



- ✓ Power consumption is minimized due to infrequently occurring very short packet transmissions with low duty cycle (<1%).
- ✓ The minimum power level defined is -3 dBm or 0.5 mW.
- ✓ Transmission, for most cases, is Line of Sight (LOS).
- ✓ Standard transmission range varies between 10m to 75m.
- ✓ Best case transmission range achieved outdoors can be upto 1000m.
- ✓ Networking topologies defined are -- Star, and Mesh.

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



IEEE 802.15.4 Variants

A/B

- Base version

C

- For China

D

- For Japan

E

- Industrial applications

F

- Active RFID uses

G

- Smart utility networks (Smart Grids)



IEEE 802.15.4 Types

Networks

Non-Beacon
Enabled

Beacon
Enabled

(a)

802.15.4

FFD

RFD

(b)

PAN
coordinator

Router

Device

Device



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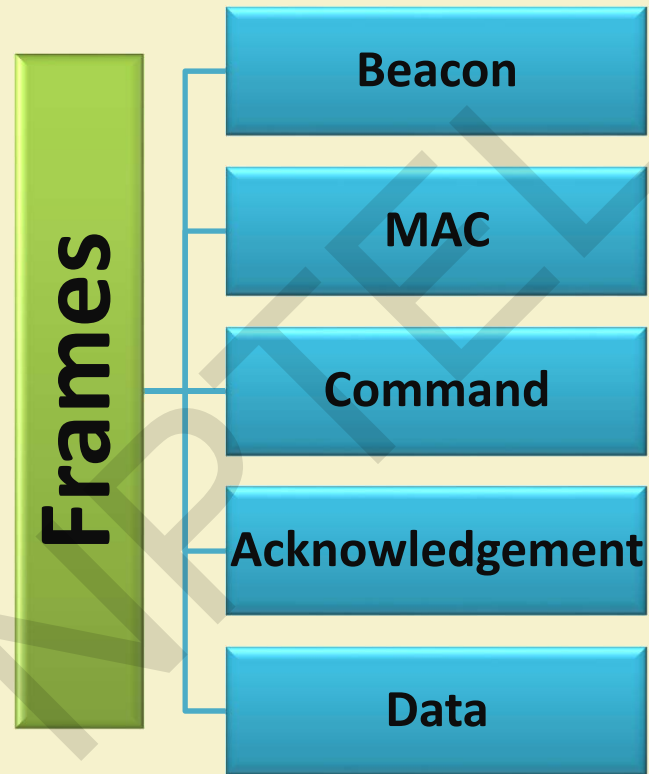
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- **Full Function Device (FFD)**
 - Can talk to all types of devices
 - Supports full protocol
- **Reduced Function Device (RFD)**
 - Can only talk to an FFD
 - Lower power consumption
 - Minimal CPU/RAM required



IEEE 802.15.4 Frames



Beacon Enabled Networks

- Periodic transmission of beacon messages
- Data-frames sent via Slotted CSMA/CA with a super frame structure managed by PAN coordinator
- Beacons used for synchronization & association of other nodes with the coordinator
- Scope of operation spans the whole network.



Non-Beacon Enabled Networks

- Data-frames sent via un-slotted CSMA/CA (Contention Based)
- Beacons used only for link layer discovery
- Requires both source and destination IDs.
- As 802.15.4 is primarily, a mesh protocol, all protocol addressing must adhere to mesh configurations
- De-centralized communication amongst nodes



Zigbee



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Features of ZigBee

- ✓ Most widely deployed enhancement of IEEE 802.15.4.
- ✓ The ZigBee protocol is defined by **layer 3 and above**. It works with the 802.15.4 layers 1 and 2.
- ✓ The standard uses layers 3 and 4 to define additional communication enhancements.
- ✓ These enhancements include authentication with valid nodes, encryption for security, and a data routing and forwarding capability that enables mesh networking.
- ✓ The most popular use of ZigBee is wireless sensor networks using the mesh topology.

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013





Important Components

ZDO

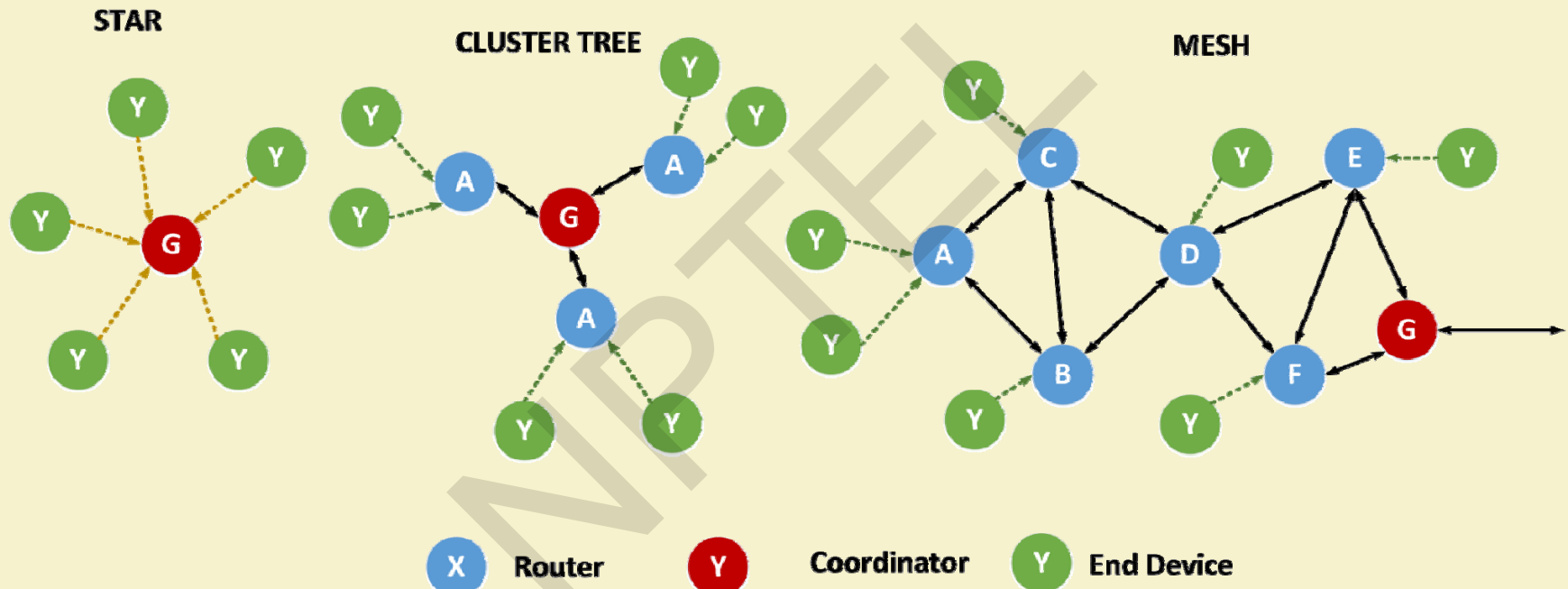
- ZigBee Device Object
(Device management, Security, Policies)

APS

- Application Support Sub-layer
(Interfacing and control services, bridge between network and other layers)



ZigBee Topologies



Source: T. Agarwal, "[ZigBee Wireless Technology Architecture and Applications](#)", Electronics Projects Focus (Online)



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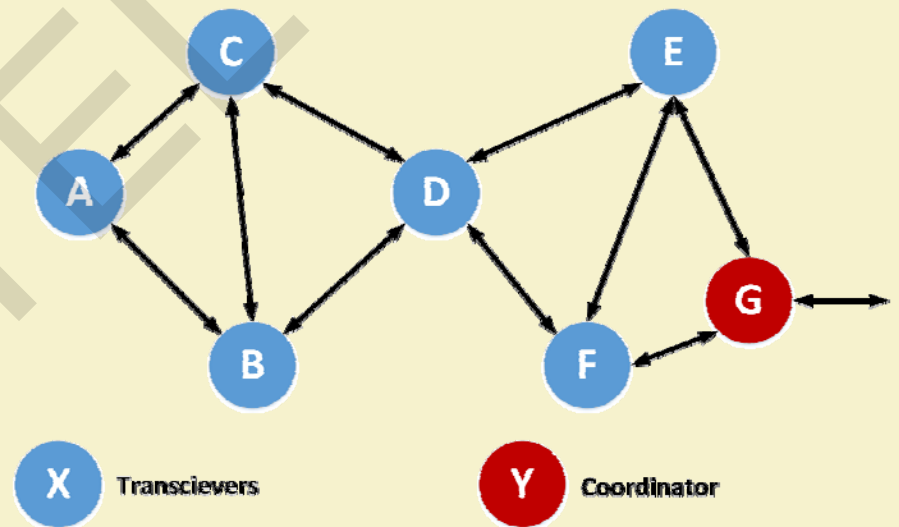


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ZigBee Mesh

- ✓ In a mesh, any node can communicate with any other node within its range.
- ✓ If nodes are not in range, messages are relayed through intermediate nodes.
- ✓ This allows the network deployment over large areas.



Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



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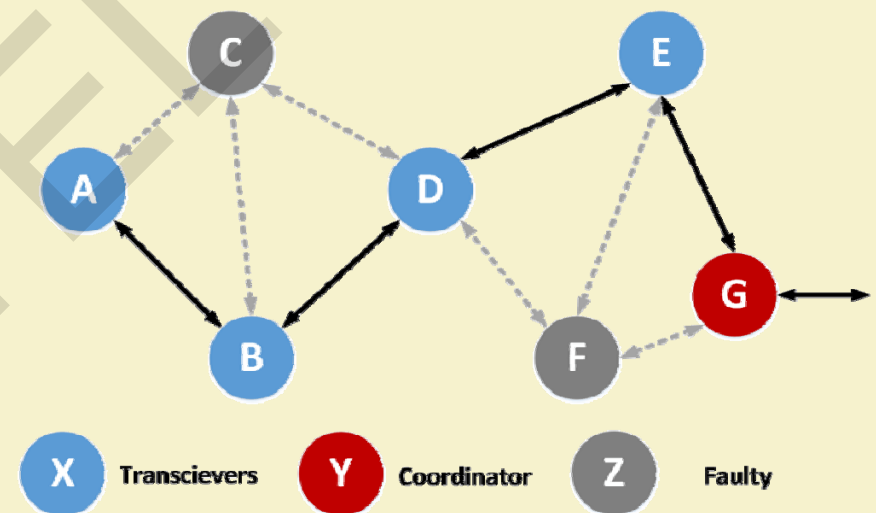


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ZigBee Mesh (Contd.)

- ✓ Meshes have increased network reliability.
- ✓ For example, if nodes C and F are down, the message packets from A can still be relayed to G via B and E.
- ✓ ZigBee mesh networks are self-configuring and self-healing.



Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



ZigBee Types

✓ *ZigBee Coordinator (ZC):*

- The Coordinator forms the root of the ZigBee network tree and might act as a bridge between networks.
- There is a single ZigBee Coordinator in each network, which originally initiates the network.
- It stores information about the network under it and outside it.
- It acts as a Trust Center & repository for security keys.

Sources:

- "[Wireless Sensor Networks Research Group](#)". [Sensor-networks.org](#). 2010-04-15.
- "[Wireless Sensor Networks Research Group](#)". [Sensor-networks.org](#). 2009-02-05.



ZigBee Types

✓ *ZigBee Router (ZR):*

- Capable of running applications, as well as relaying information between nodes connected to it.

✓ *ZigBee End Device (ZED):*

- It contains just enough functionality to talk to the parent node, and it cannot relay data from other devices.
- This allows the node to be asleep a significant amount of the time thereby enhancing battery life.
- Memory requirements and cost of ZEDs are quite low, as compared to ZR or ZC.

Sources:

- "[Wireless Sensor Networks Research Group](#)". [Sensor-networks.org](#). 2010-04-15.
- "[Wireless Sensor Networks Research Group](#)". [Sensor-networks.org](#). 2009-02-05.



ZigBee Network Layer

- ✓ The network layer uses Ad Hoc On-Demand Distance Vector (AODV) routing.
- ✓ To find the final destination, the AODV broadcasts a route request to all its immediate neighbors.
- ✓ The neighbors relay the same information to their neighbors, eventually spreading the request throughout the network.
- ✓ Upon discovery of the destination, a low-cost path is calculated and informed to the requesting device via unicast messaging.

Source: ["Zigbee", Wikipedia \(Online\)](#)



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Applications

- ✓ Building automation
- ✓ Remote control (RF4CE or RF for consumer electronics)
- ✓ Smart energy for home energy monitoring
- ✓ Health care for medical and fitness monitoring
- ✓ Home automation for control of smart homes
- ✓ Light Link for control of LED lighting
- ✓ Telecom services

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



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Sensing

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Definition

- ✓ A sensor detects (senses) changes in the ambient conditions or in the state of another device or a system, and forwards or processes this information in a certain manner [1].

“A device which detects or measures a physical property and records, indicates, or otherwise responds to it” [2].

- Oxford Dictionary

References:

1. <http://www.businessdictionary.com/definition/sensor.html>
2. <https://en.oxforddictionaries.com/definition/sensor>



Sensors

- ✓ They perform some input functions by sensing or feeling the physical changes in characteristics of a system in response to a stimuli.
- ✓ For example heat is converted to electrical signals in a temperature sensor, or atmospheric pressure is converted to electrical signals in a barometer.



Transducers

- ✓ Transducers convert or transduce energy of one kind into another.
- ✓ For example, in a sound system, a microphone (input device) converts sound waves into electrical signals for an amplifier to amplify (a process), and a loudspeaker (output device) converts these electrical signals back into sound waves.



Sensor vs. Transducer

- ✓ The word “Transducer” is the collective term used for both **Sensors** which can be used to sense a wide range of different energy forms such as movement, electrical signals, radiant energy, thermal or magnetic energy etc., and **Actuators** which can be used to switch voltages or currents [1].

References:

1. http://www.electronics-tutorials.ws/io/io_1.html



Sensor Features

- ✓ It is only sensitive to the measured property (e.g., A temperature sensor senses the ambient temperature of a room.)
- ✓ It is insensitive to any other property likely to be encountered in its application (e.g., A temperature sensor does not bother about light or pressure while sensing the temperature.)
- ✓ It does not influence the measured property (e.g., measuring the temperature does not reduce or increase the temperature).

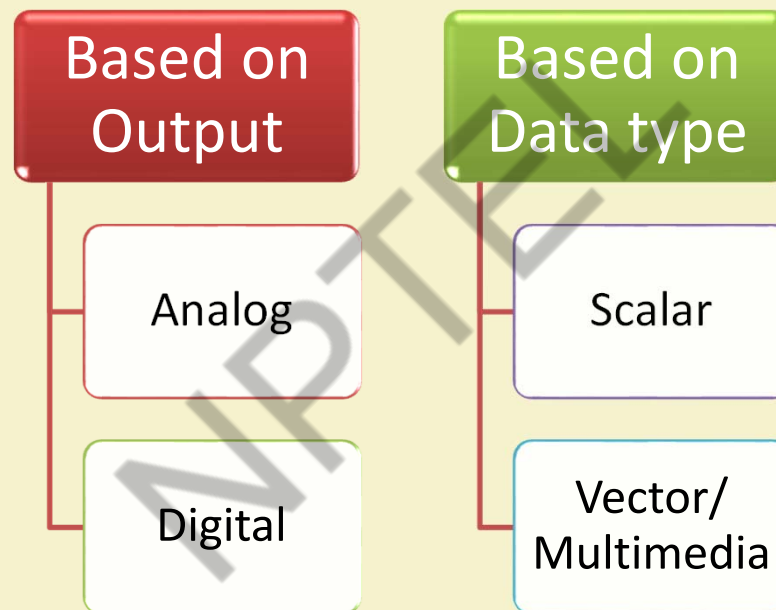


Sensor Resolution

- ✓ The resolution of a sensor is the smallest change it can detect in the quantity that it is measuring.
- ✓ The resolution of a sensor with a digital output is usually the smallest resolution the digital output it is capable of processing.
- ✓ The more is the resolution of a sensor, the more accurate is its precision.
- ✓ A sensor's accuracy does not depend upon its resolution.



Sensor Classes



Analog Sensors

- ✓ **Analog Sensors** produce a continuous output signal or voltage which is generally proportional to the quantity being measured.
- ✓ Physical quantities such as Temperature, Speed, Pressure, Displacement, Strain etc. are all analog quantities as they tend to be continuous in nature.
- ✓ For example, the temperature of a liquid can be measured using a thermometer or thermocouple (e.g. in geysers) which continuously responds to temperature changes as the liquid is heated up or cooled down.



Digital Sensors

- ✓ **Digital Sensors** produce discrete digital output signals or voltages that are a digital representation of the quantity being measured.
- ✓ Digital sensors produce a binary output signal in the form of a logic “1” or a logic “0”, (“ON” or “OFF”).
- ✓ Digital signal only produces discrete (non-continuous) values, which may be output as a single “bit” (serial transmission), or by combining the bits to produce a single “byte” output (parallel transmission).



Scalar Sensors

- ✓ **Scalar Sensors** produce output signal or voltage which is generally proportional to the magnitude of the quantity being measured.
- ✓ Physical quantities such as temperature, color, pressure, strain, etc. are all scalar quantities as only their magnitude is sufficient to convey an information.
- ✓ For example, the temperature of a room can be measured using a thermometer or thermocouple, which responds to temperature changes irrespective of the orientation of the sensor or its direction.



Vector Sensors

- ✓ **Vector Sensors** produce output signal or voltage which is generally proportional to the magnitude, direction, as well as the orientation of the quantity being measured.
- ✓ Physical quantities such as sound, image, velocity, acceleration, orientation, etc. are all vector quantities, as only their magnitude is not sufficient to convey the complete information.
- ✓ For example, the acceleration of a body can be measured using an accelerometer, which gives the components of acceleration of the body with respect to the x,y,z coordinate axes.



Sensor Types

Light

- Light Dependent resistor
- Photo-diode

Temperature

- Thermocouple
- Thermistor

Force

- Strain gauge
- Pressure switch

Position

- Potentiometer, Encoders
- Opto-coupler

Speed

- Reflective/ Opto-coupler
- Doppler effect sensor

Sound

- Carbon Microphone
- Piezoelectric Crystal

Chemical

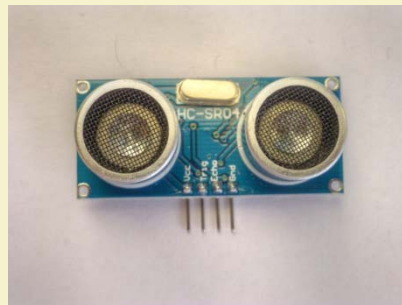
- Liquid Chemical sensor
- Gaseous chemical sensor





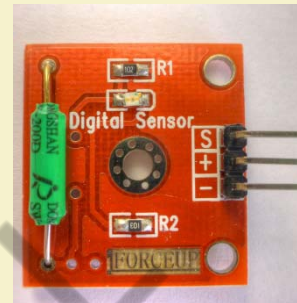
Pressure Sensor

Source: Wikimedia Commons



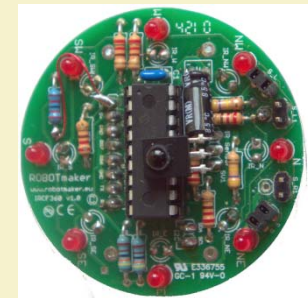
Ultrasonic Distance Sensor

Source: Wikimedia Commons



Tilt Sensor

Source: Wikimedia Commons



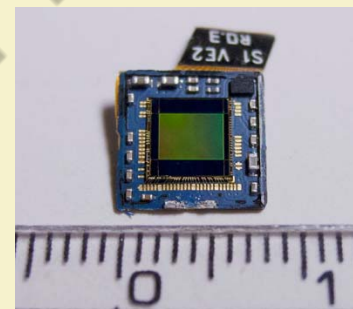
Infrared Motion Sensor

Source: Wikimedia Commons



Analog Temperature Sensor

Source: Wikimedia Commons



Camera Sensor

Source: Wikimedia Commons



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Sensorial Deviations

- ✓ Since the range of the output signal is always limited, the output signal will eventually reach a minimum or maximum, when the measured property exceeds the limits. The full scale range of a sensor defines the maximum and minimum values of the measured property.
- ✓ The sensitivity of a sensor under real conditions may differ from the value specified. This is called a **sensitivity error**.
- ✓ If the output signal differs from the correct value by a constant, the sensor has an **offset error** or **bias**.

Reference: <https://en.wikipedia.org/wiki/Sensor>



Non-linearity

- ✓ Nonlinearity is deviation of a sensor's transfer function (TF) from a straight line transfer function.
- ✓ This is defined by the amount the output differs from ideal TF behavior over the full range of the sensor, which is denoted as the percentage of the full range.
- ✓ Most sensors have linear behavior.

Reference: <https://en.wikipedia.org/wiki/Sensor>



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- ✓ If the output signal slowly changes independent of the measured property, this is defined as **drift**. Long term drift over months or years is caused by physical changes in the sensor.
- ✓ **Noise** is a random deviation of the signal that varies in time.

Reference: <https://en.wikipedia.org/wiki/Sensor>



Hysteresis Error

- ✓ A hysteresis error causes the sensor output value to vary depending on the sensor's previous input values.
- ✓ If a sensor's output is different depending on whether a specific input value was reached by increasing or decreasing the input, then the sensor has a hysteresis error.
- ✓ The present reading depends on the past input values.
- ✓ Typically in analog sensors, magnetic sensors, heating of metal strips.

Reference: <https://en.wikipedia.org/wiki/Sensor>



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Other Errors

- ✓ If the sensor has a digital output, the output is essentially an approximation of the measured property. This error is also called **quantization error**.
- ✓ If the signal is monitored digitally, the sampling frequency can cause a dynamic error, or if the input variable or added noise changes periodically at a frequency proportional to the multiple of the sampling rate, **aliasing errors** may occur.
- ✓ The sensor may to some extent be sensitive to properties other than the property being measured. For example, most sensors are influenced by the temperature of their environment.

Reference: <https://en.wikipedia.org/wiki/Sensor>



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Actuation

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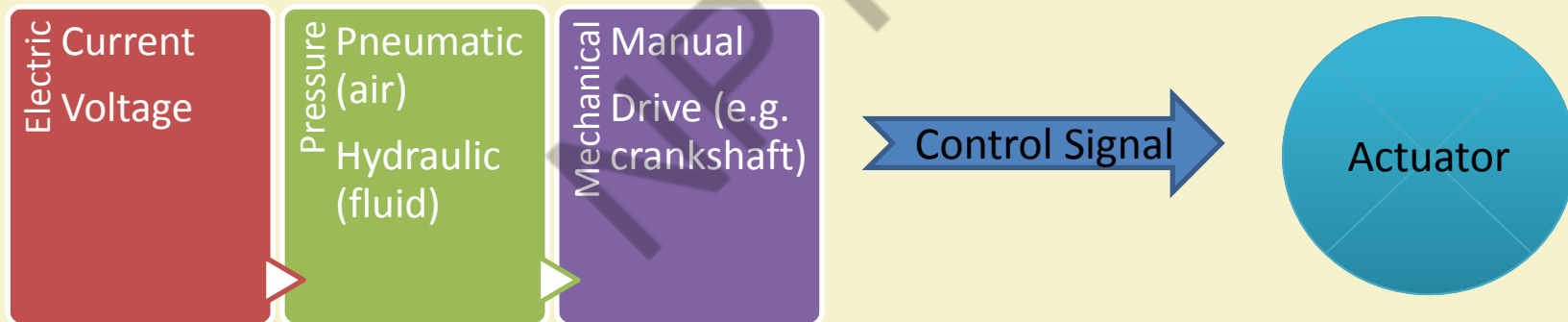
Website: <http://cse.iitkgp.ac.in/~smisra/>

Actuator

- ✓ An actuator is a component of a machine or system that moves or controls the mechanism or the system.
- ✓ An actuator is the mechanism by which a control system acts upon an environment
- ✓ An actuator requires a control signal and a source of energy.



- ✓ Upon receiving a control signal is received, the actuator responds by converting the energy into mechanical motion.
- ✓ The control system can be simple (a fixed mechanical or electronic system), software-based (e.g. a printer driver, robot control system), a human, or any other input.



Actuator Types

Hydraulic

Pneumatic

Electrical

Thermal/ Magnetic

Mechanical



Hydraulic Actuators

- ✓ A hydraulic actuator consists of a cylinder or fluid motor that uses hydraulic power to facilitate mechanical operation.
- ✓ The mechanical motion is converted to linear, rotary or oscillatory motion.
- ✓ Since liquids are nearly impossible to compress, a hydraulic actuator exerts considerable force.
- ✓ The actuator's limited acceleration restricts its usage.

Reference: <https://en.wikipedia.org/wiki/Actuator>



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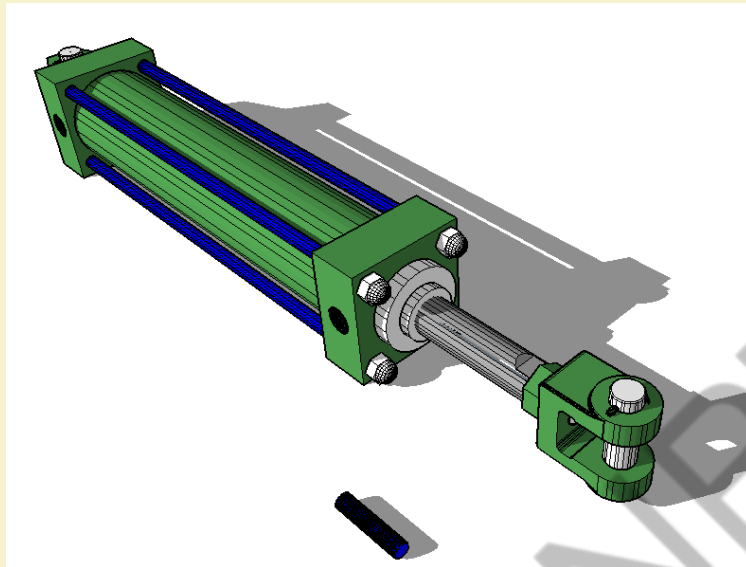


Fig: An oil based hydraulic actuator

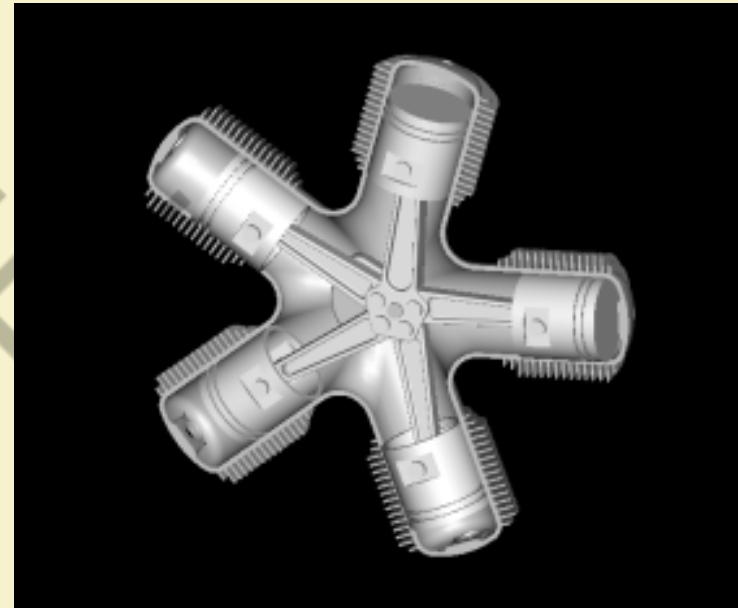


Fig: A radial engine acts as a hydraulic actuator

Source: Wikimedia Commons

File: Radial_engine.gif



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Pneumatic Actuators

- ✓ A pneumatic actuator converts energy formed by vacuum or compressed air at high pressure into either linear or rotary motion.
- ✓ Pneumatic rack and pinion actuators are used for valve controls of water pipes.
- ✓ Pneumatic energy quickly responds to starting and stopping signals.
- ✓ The power source does not need to be stored in reserve for operation.

Reference: <https://en.wikipedia.org/wiki/Actuator>



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- ✓ Pneumatic actuators enable large forces to be produced from relatively small pressure changes (e.g., Pneumatic brakes can are very responsive to small changes in pressure applied by the driver).
- ✓ It is responsible for converting pressure into force.



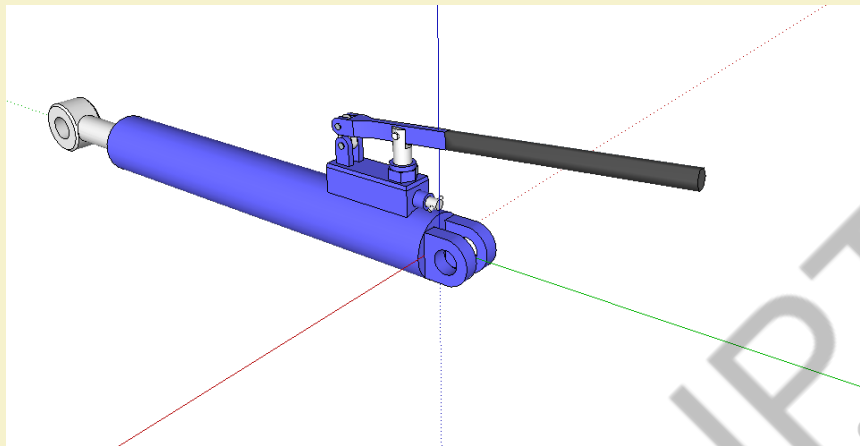


Fig: A manual linear pneumatic actuator

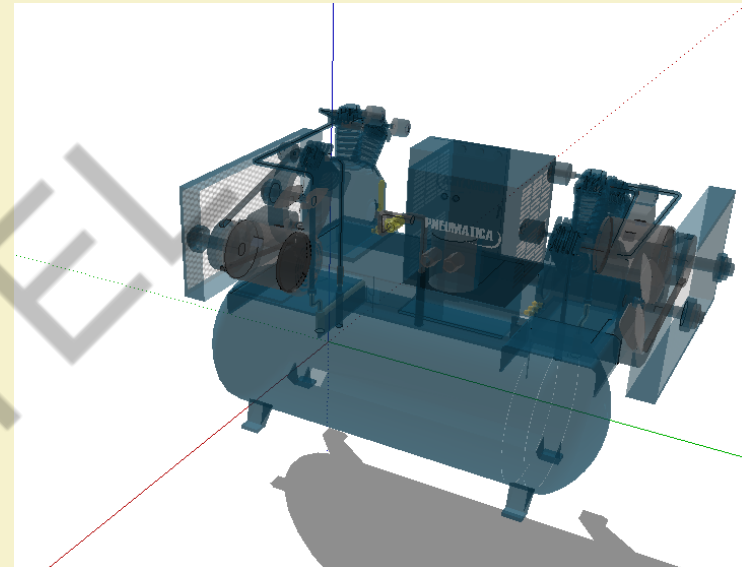


Fig: An air pump acts as a pneumatic actuator



Electric Actuators

- ✓ An electric actuator is generally powered by a motor that converts electrical energy into mechanical torque.
- ✓ The electrical energy is used to actuate equipment such as solenoid valves which control the flow of water in pipes in response to electrical signals.
- ✓ Considered as one of the cheapest, cleanest and speedy actuator types available.

Reference: <https://en.wikipedia.org/wiki/Actuator>



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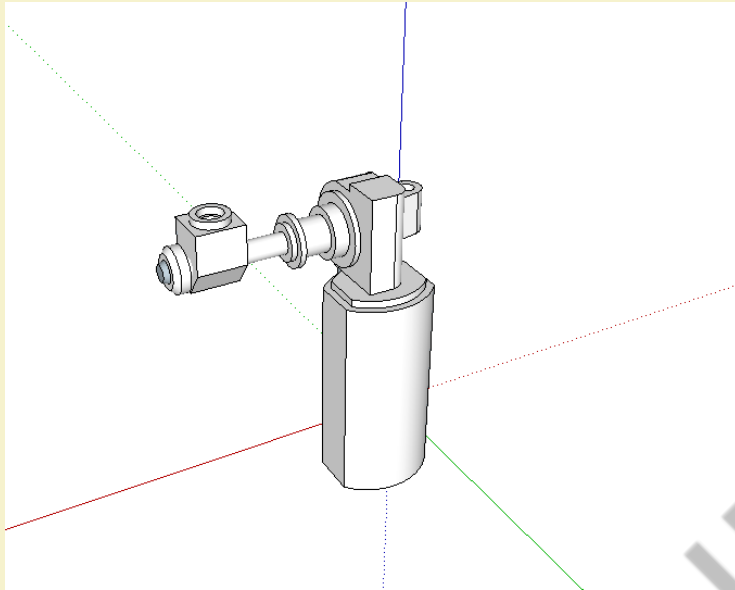


Fig: A motor drive-based rotary actuator

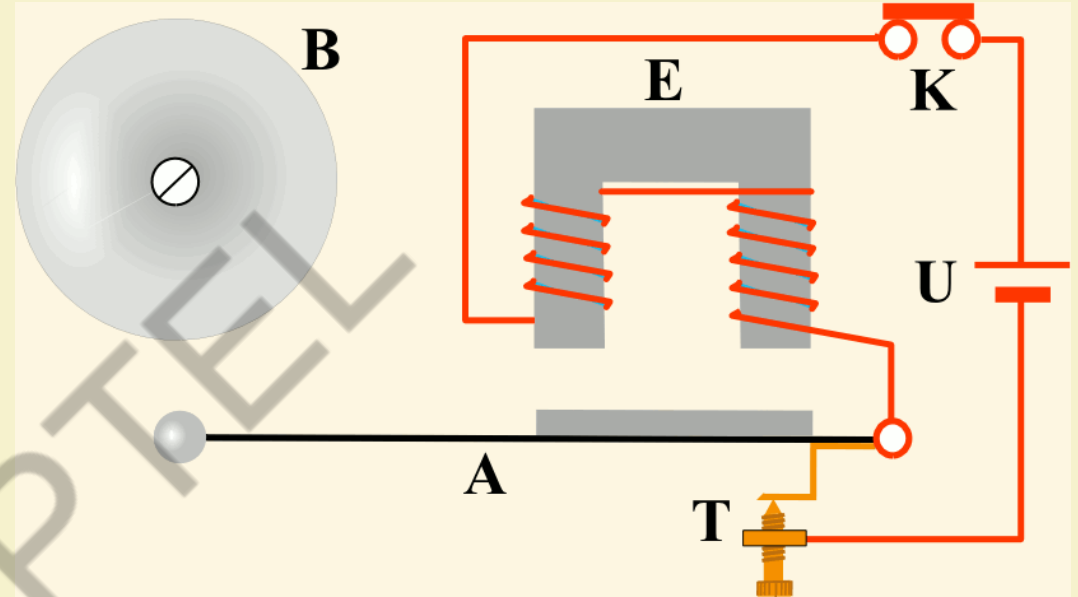


Fig: A solenoid based electric bell ringing mechanism

Source: Wikimedia Commons

File: Electric_Bell_animation.gif



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Thermal or Magnetic Actuators

- ✓ These can be actuated by applying thermal or magnetic energy.
- ✓ They tend to be compact, lightweight, economical and with high power density.
- ✓ These actuators use shape memory materials (SMMs), such as shape memory alloys (SMAs) or magnetic shape-memory alloys (MSMAs).
- ✓ Some popular manufacturers of these devices are *Finnish Modti Inc.* and *American Dynalloy*.

Reference: <https://en.wikipedia.org/wiki/Actuator>



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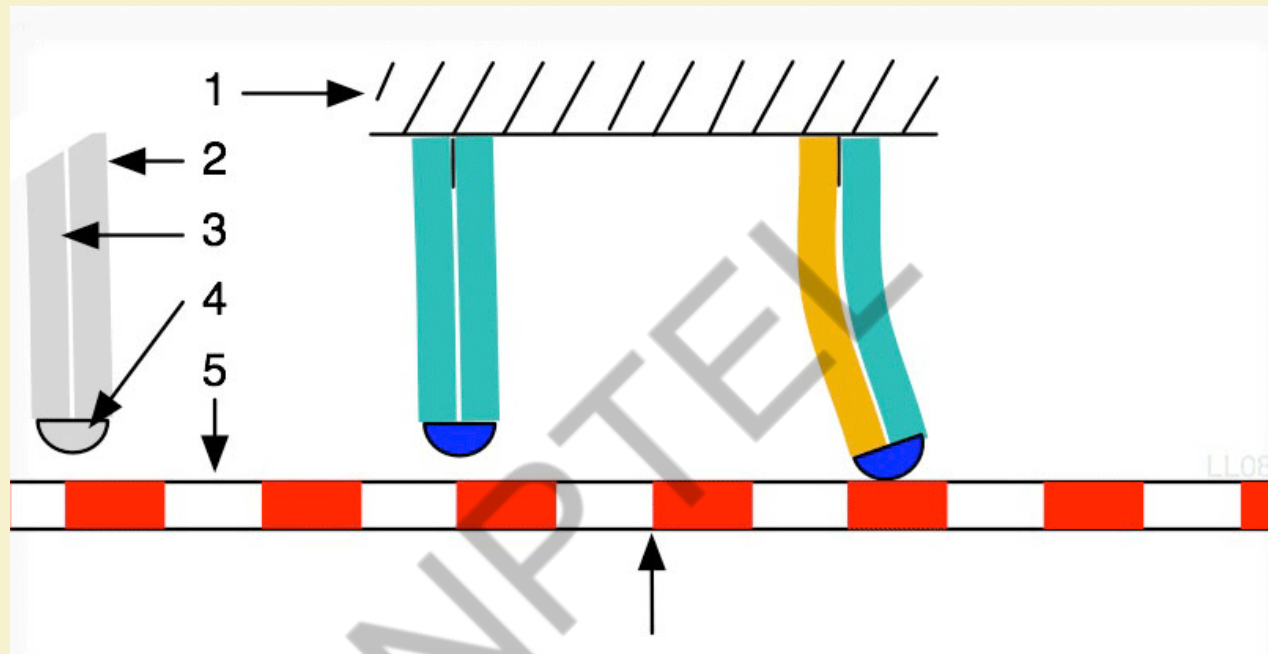


Fig: A piezo motor using SMA

Source: Wikimedia Commons

File: Piezomotor type bimorph.gif



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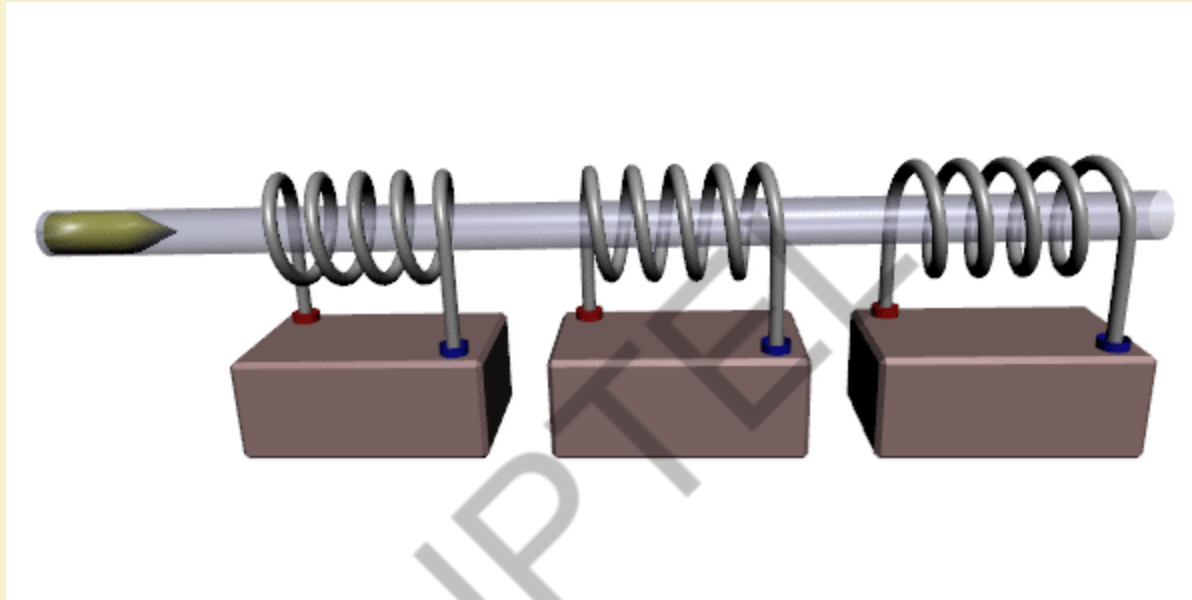


Fig: A coil gun works on the principle of magnetic actuation

Source: Wikimedia Commons

File: Coilgun animation.gif



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Mechanical Actuators

- ✓ A mechanical actuator converts rotary motion into linear motion to execute some movement.
- ✓ It involves gears, rails, pulleys, chains and other devices to operate.
- ✓ Example: rack and pinion.



Fig: A rack and pinion mechanism

Source: Wikimedia Commons

File: Rack and pinion.png

Reference: <https://en.wikipedia.org/wiki/Actuator>



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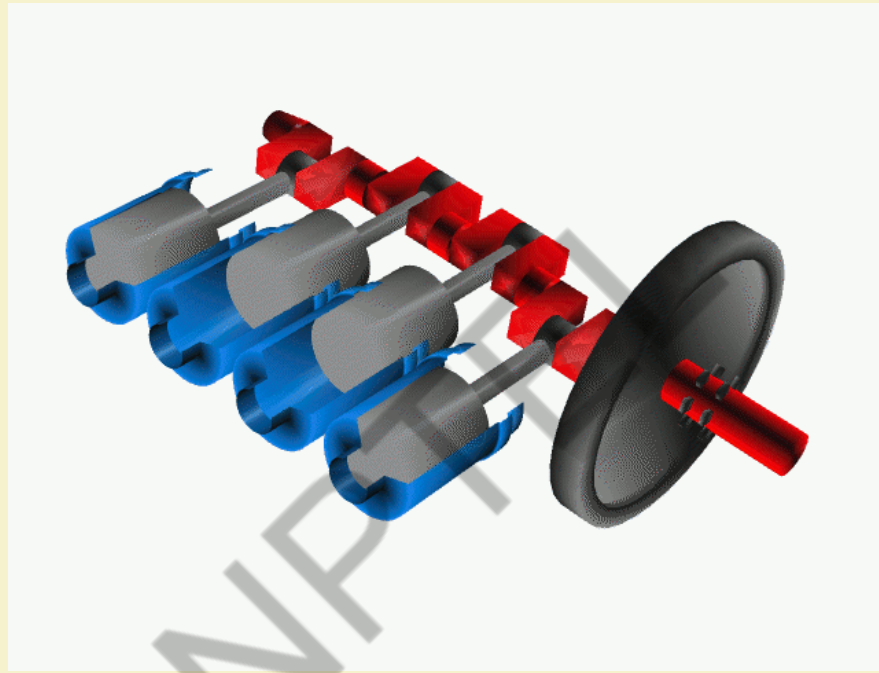


Fig: A crank shaft acting as a mechanical actuator

Source: Wikimedia Commons

File: Cshaft.gif



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Soft Actuators

- ✓ Soft actuators (e.g. polymer based) are designed to handle fragile objects like fruit harvesting in agriculture or manipulating the internal organs in biomedicine.
- ✓ They typically address challenging tasks in robotics.
- ✓ Soft actuators produce flexible motion due to the integration of microscopic changes at the molecular level into a macroscopic deformation of the actuator materials.

Reference: <https://en.wikipedia.org/wiki/Actuator>



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Shape Memory Polymers

- ✓ Shape memory polymer (SMP) actuators function similar to our muscles, even providing a response to a range of stimuli such as light, electrical, magnetic, heat, pH, and moisture changes.
- ✓ SMP exhibits surprising features such a low density, high strain recovery, biocompatibility, and biodegradability.

Reference: <https://en.wikipedia.org/wiki/Actuator>



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Light Activated Polymers

- ✓ Photopolymer/light activated polymers (LAP) are a special type of SMP that are activated by light stimuli.
- ✓ The LAP actuators have instant response.
- ✓ They can be controlled remotely without any physical contact, only using the variation of light frequency or intensity.

Reference: <https://en.wikipedia.org/wiki/Actuator>



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Basics of IoT Networking – Part I

Dr. Sudip Misra

Associate Professor

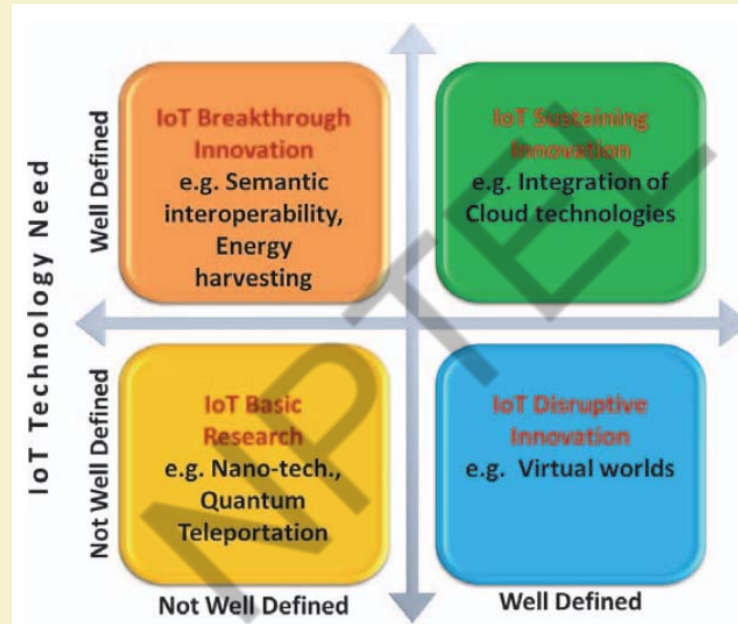
Department of Computer Science and Engineering

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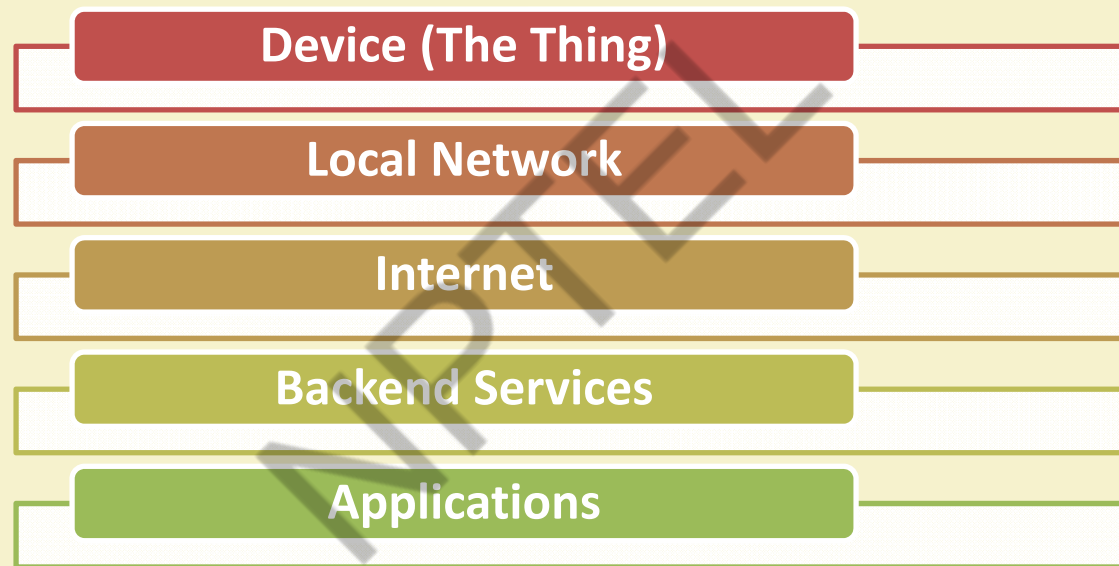
Convergence of Domains

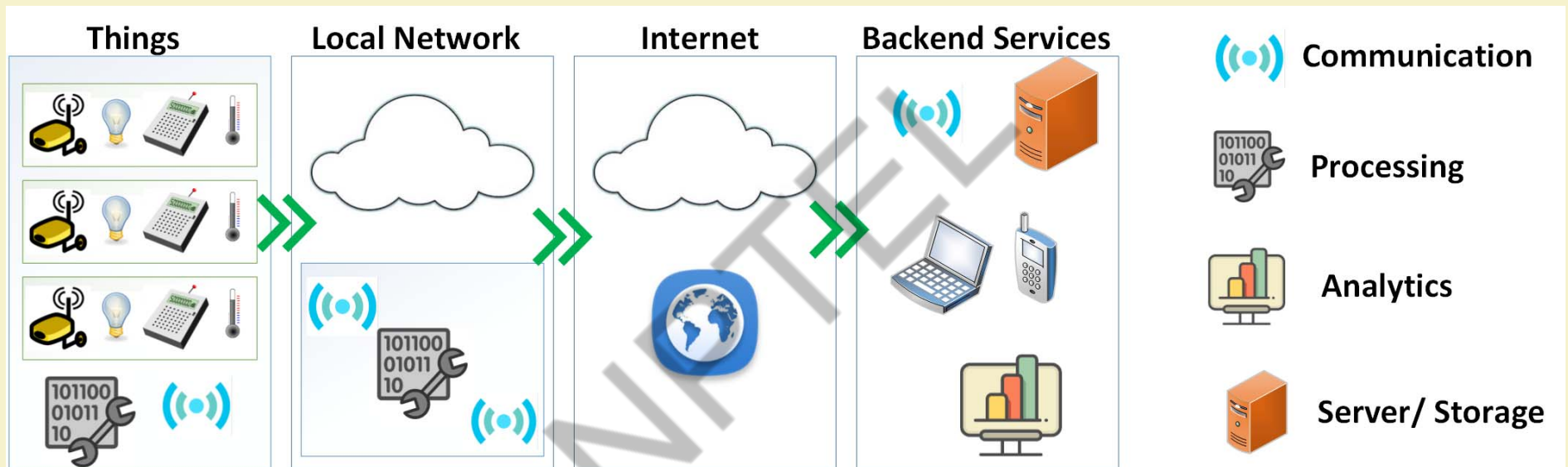


Source: O. Vermesan, P. Friess, "Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems", River Publishers, Series in Communications, 2013



IoT Components





Functional Components of IoT

- ✓ Component for interaction and communication with other IoT devices
- ✓ Component for processing and analysis of operations
- ✓ Component for Internet interaction
- ✓ Components for handling Web services of applications
- ✓ Component to integrate application services
- ✓ User interface to access IoT

Source: O Vermesan, P. Friess, “Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers, Series in Communications, 2013



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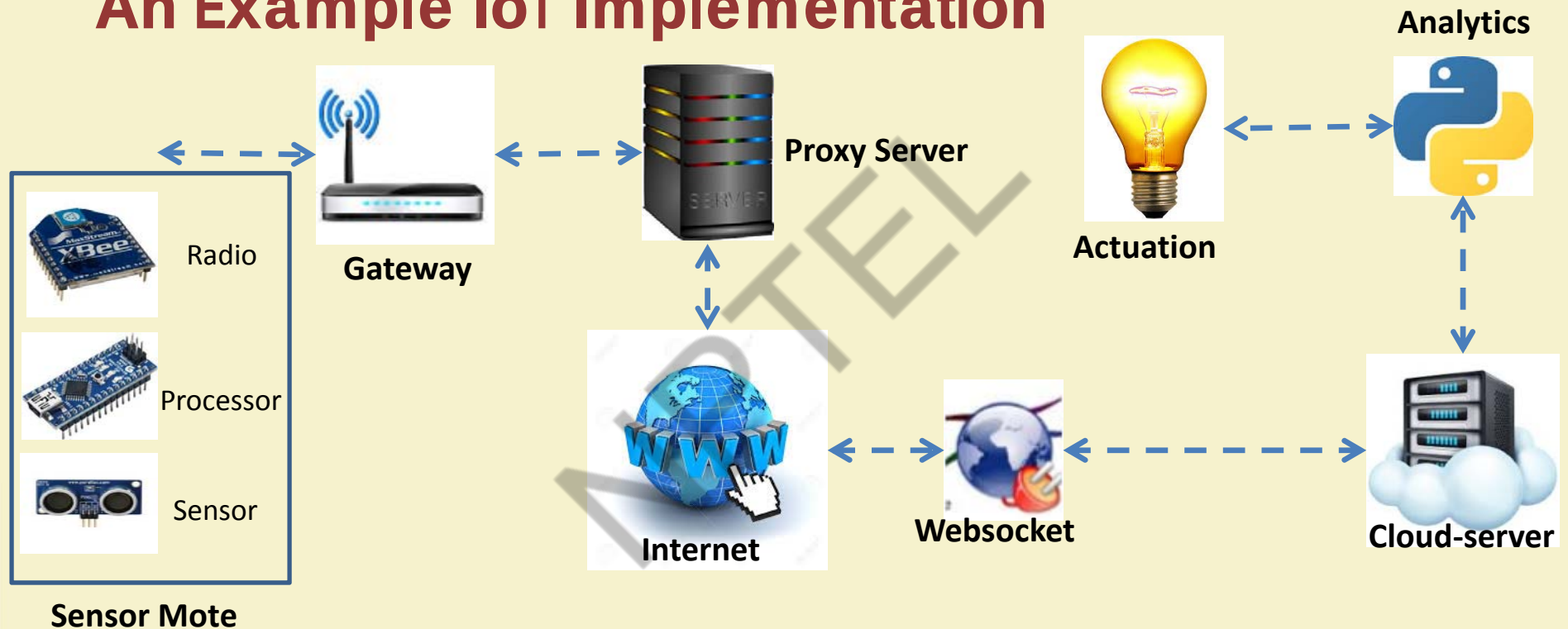


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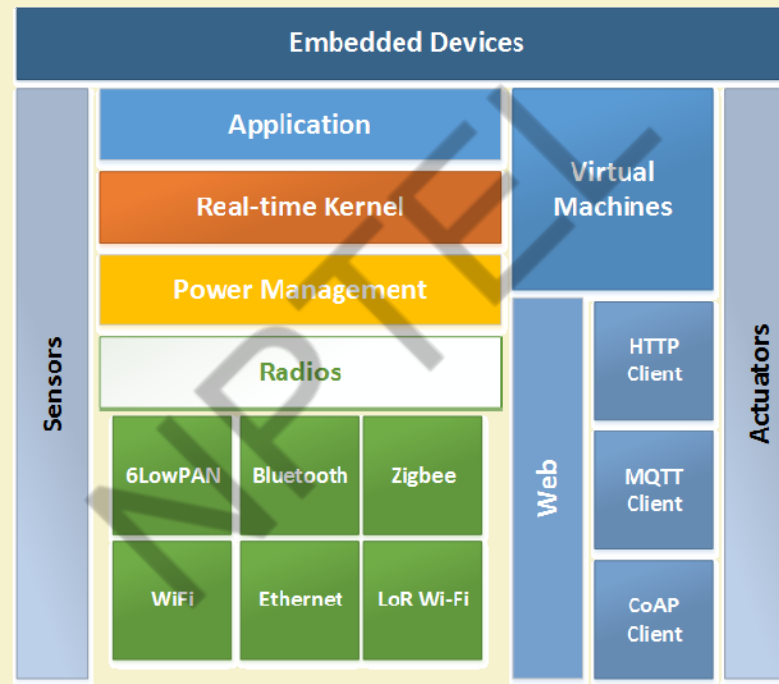
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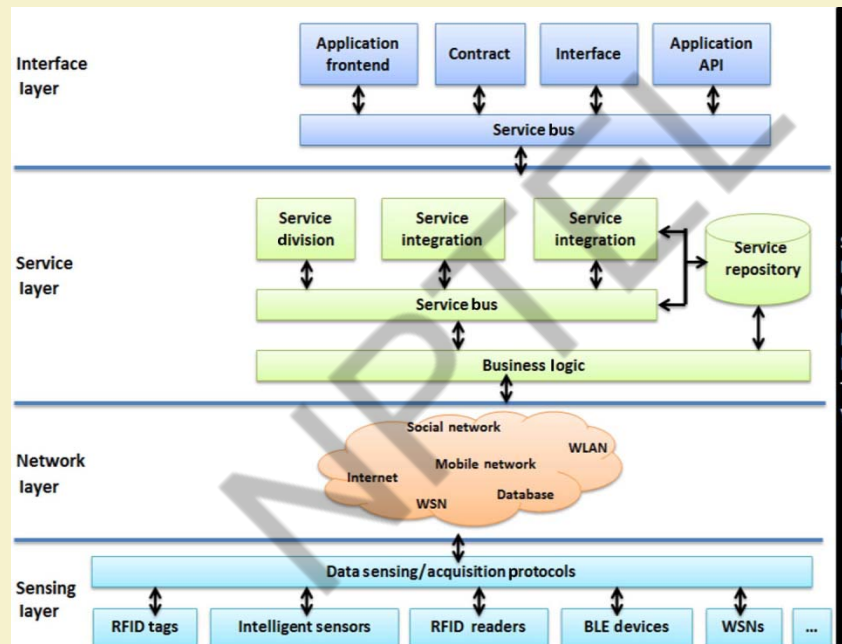
An Example IoT Implementation



IoT Interdependencies



IoT Service Oriented Architecture



Source: Li Da Xu, Wu He, and Shancang Li, "Internet of Things in Industries: A Survey ", IEEE Transactions on Industrial Informatics, Vol. 10, No. 4, Nov. 2014.



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8

IoT Categories

✓ Industrial IoT

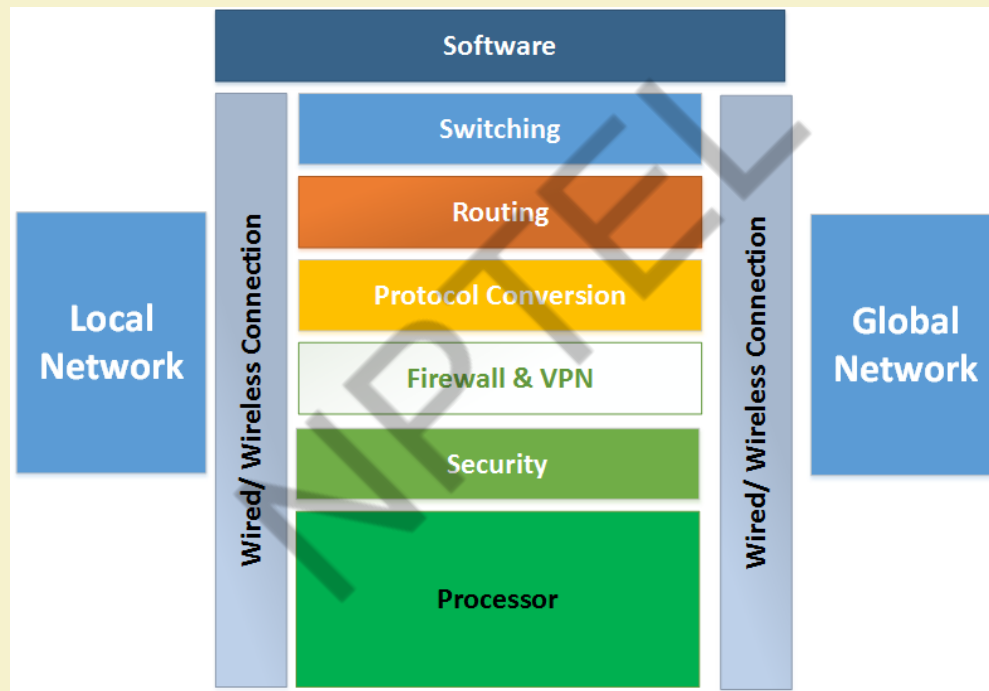
- IoT device connects to an IP network and the global Internet.
- Communication between the nodes done using regular as well as industry specific technologies.

✓ Consumer IoT

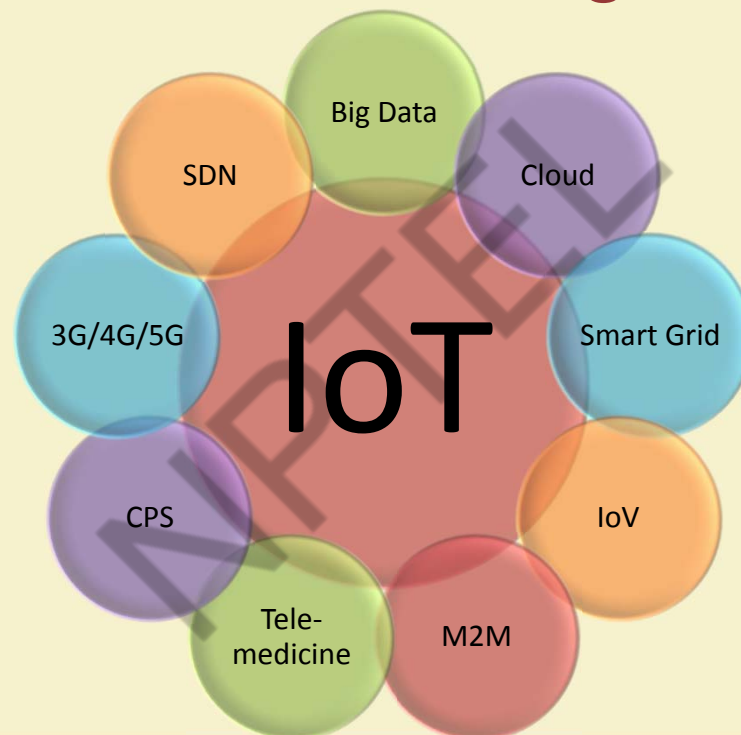
- IoT device communicates within the locally networked devices.
- Local communication is done mainly via Bluetooth, Zigbee or WiFi.
- Generally limited to local communication by a Gateway



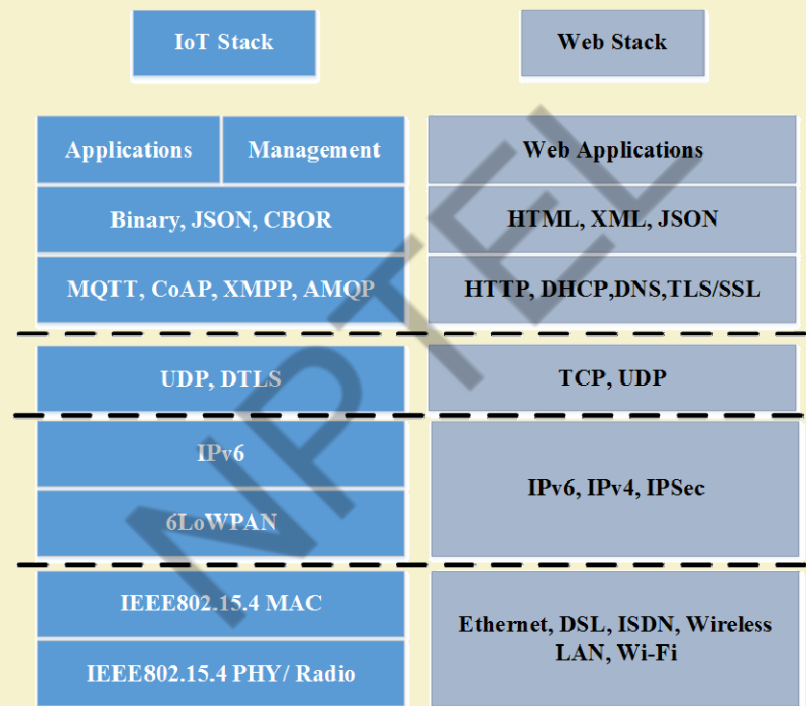
IoT Gateways



IoT and Associated Technologies



Technical Deviations from Regular Web



Key Technologies for IoT



Source: O Vermesan, P. Friess, "Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems", River Publishers, Series in Communications, 2013



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IoT Challenges

- ✓ Security
- ✓ Scalability
- ✓ Energy efficiency
- ✓ Bandwidth management
- ✓ Modeling and Analysis
- ✓ Interfacing
- ✓ Interoperability
- ✓ Data storage
- ✓ Data Analytics
- ✓ Complexity management (e.g., SDN)



Considerations

- ✓ Communication between the IoT device(s) and the outside world dictates the network architecture.
- ✓ Choice of communication technology dictates the IoT device hardware requirements and costs.
- ✓ Due to the presence of numerous applications of IoT enabled devices, a single networking paradigm not sufficient to address all the needs of the consumer or the IoT device.



Complexity of Networks

- ✓ Growth of networks
- ✓ Interference among devices
- ✓ Network management
- ✓ Heterogeneity in networks
- ✓ Protocol standardization within networks

Source: O Vermesan, P. Friess, “Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers, Series in Communications, 2013



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Wireless Networks

- Traffic and load management
- Variations in wireless networks – Wireless Body Area Networks and other Personal Area Networks
- Interoperability
- Network management
- Overlay networks

Source: O. Vermesan, P. Friess, “Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers, Series in Communications, 2013



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Scalability

- Flexibility within Internet
- IoT integration
- Large scale deployment
- Real-time connectivity of billions of devices



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Introduction to IoT – Part II

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IoT Resulting in Address Crunch

- ✓ Estimated 20-50 billion devices by 2018
- ✓ Reason is the integration of existing devices, smart devices as well as constrained nodes in a singular framework.
- ✓ Integration of various connectivity features such as cellular, Wi-Fi, ethernet with upcoming ones such as Bluetooth Low Energy (BLE), DASH7, Insteon, IEEE 802.15.4, etc.
- ✓ The ITU vision is approaching reality as the present day networked devices have outnumbered humans on earth.

Reference:

- ✓ Cisco Systems, (2011). *The Internet of Things How the Next Evolution of the Internet Is Changing Everything* [Online]. Available: http://www.cisco.com/web/about/ac79/docs/innov/IoT_IBSG_0411FINAL.pdf
- ✓ ITU Broadband Commission, (2012). *The State of Broadband 2012: Achieving Digital Inclusion for All ITU Broadband Commission Report*, [Online]. Available: <http://www.broadbandcommission.org/Documents/bbannualreport2012.pdf>
- ✓ Ericsson, (2011). *More than 50 Billion Connected Devices*, [Online]. Available: <http://www.ericsson.com/res/docs/whitepapers/wp-50billions.pdf>

Connectivity Terminologies

IoT LAN

- Local, Short range Comm, May or may not connect to Internet, Building or Organization wide

IoT WAN

- Connection of various network segments, Organizationally and geographically wide, Connects to the internet

IoT Node

- Connected to other nodes inside a LAN via the IoT LAN, May be sometimes connected to the internet through a WAN directly

IoT Gateway

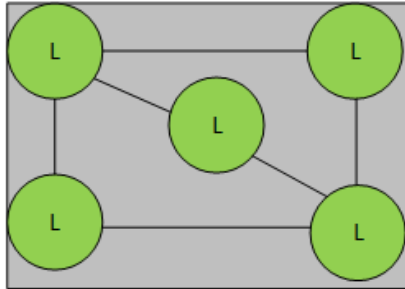
- A router connecting the IoT LAN to a WAN to the Internet, Can implement several LAN and WAN, Forwards packets between LAN and WAN on the IP layer

IoT Proxy

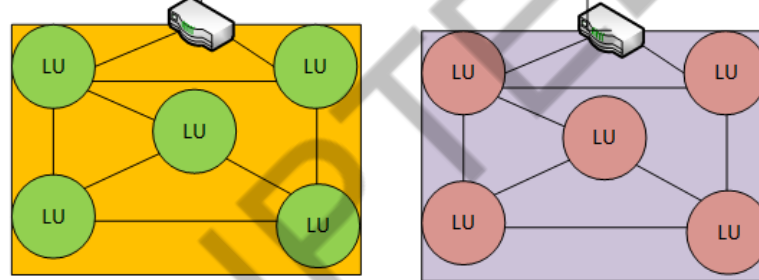
- Performs active application layer functions between IoT nodes and other entities

IoT Network Configurations

IoT LAN



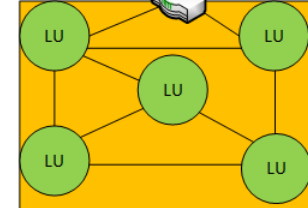
IoT WAN



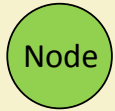
Internet



IoT Proxy



Gateway

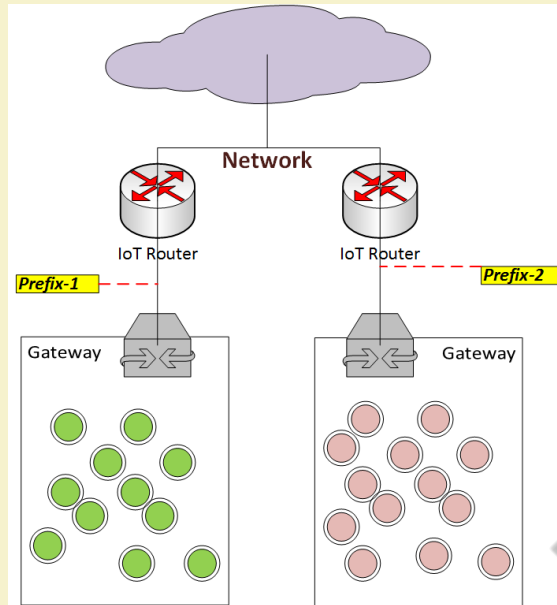


Node

Source: Teemu Savolainen, Jonne Soininen, and Bilhanan Silverajan, "IPv6 Addressing Strategies for IoT", IEEE Sensors Journal, Vol. 13, No. 10, Oct 2013

- ✓ Some of the IoT network configurations restricted to local areas, analogous to normal LANs, WANs and proxy are shown in the previous figures.
- ✓ The nodes represented by green circles have **L**: local link addresses or **LU**: local link addresses which are unique locally.
- ✓ Nodes within a gateway's jurisdiction have addresses that are valid within the gateway's domain only.
- ✓ The same addresses may be repeated in the domain of another gateway. The gateway has a unique network prefix, which can be used to identify them globally.
- ✓ This strategy saves a lot of unnecessary address wastage. Although, the nodes have to communicate to the internet via the gateway.

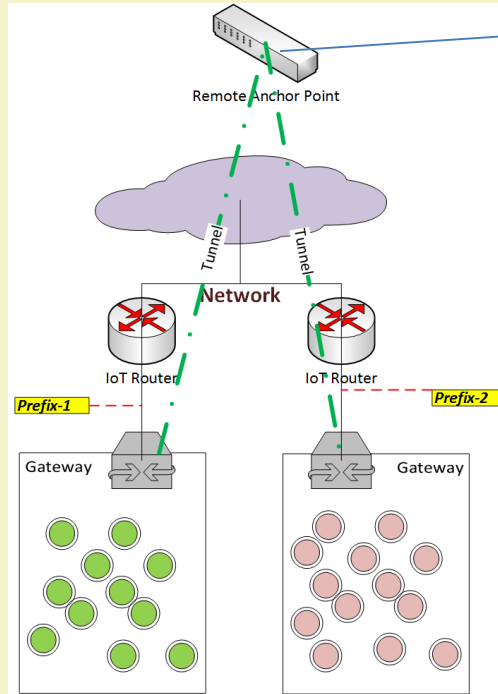
Gateway Prefix Allotment



- ✓ One of the strategies of address conservation in IoT is to use local addresses which exist uniquely within the domain of the gateway. These are represented by the circles in this slide.
- ✓ The network connected to the internet has routers with their set of addresses and ranges.
- ✓ These routers have multiple gateways connected to them which can forward packets from the nodes, to the Internet, only via these routers. These routers assign prefixes to gateways under them, so that the gateways can be identified with them.

Source: Teemu Savolainen, Jonne Soininen, and Bilhanan Silverajan, "IPv6 Addressing Strategies for IoT", IEEE Sensors Journal, Vol. 13, No. 10, Oct 2013

Impact of Mobility on Addressing



Has the
global view
of the
network
underneath

- ✓ The network prefix changes from 1 to 2 due to movement, making the IoT LAN safe from changes due to movements.
- ✓ IoT gateway WAN address changes without change in LAN address. This is achieved using ULA.

Source: Teemu Savolainen, Jonne Soininen, and Bilhanan Silverajan, "IPv6 Addressing Strategies for IoT", IEEE Sensors Journal, Vol. 13, No. 10, Oct 2013

- ✓ The gateways assigned with prefixes, which are attached to a remote anchor point by using various protocols such as Mobile IPv6, and are immune to changes of network prefixes.
- ✓ This is achieved using LU. The address of the nodes within the gateways remain unchanged as the gateways provide them with locally unique address and the change in gateway's network prefix doesn't affect them.
- ✓ Sometimes, there is a need for the nodes to communicate directly to the internet. This is achieved by tunneling, where the nodes communicate to a remote anchor point instead of channeling their packets through the router which is achieved by using tunneling protocols such as IKEv2: internet key exchange version 2

Gateways

- ✓ IoT gateways with or without proxies responsible mainly for:
 - Internet connectivity
 - IoT LAN intra-connectivity
- ✓ Upstream address prefixes are obtained using mechanisms like DHCPv6 and delegated to the nodes using SLAAC (stateless addressing).
- ✓ LU addresses are maintained independently of globally routable addresses, in cases where internal address stability is of prime concern.

Source: Teemu Savolainen, Jonne Soininen, and Bilhanan Silverajan, "IPv6 Addressing Strategies for IoT", IEEE Sensors Journal, Vol. 13, No. 10, Oct 2013

- ✓ Despite providing address stability, LU cannot communicate directly with the internet or the upper layers, which is solved by implementing an application layer proxy.
- ✓ Application layer proxies may be additionally configured to process data, rather than just passing it.
- ✓ In nodes with no support for computationally intensive tasks, IoT proxy gathers data sent to the link-local multicast address and routes them globally.

- ✓ Presently, the Internet is mainly IPv4, based with little or no IPv6 uplink facilities or support.
- ✓ Due to the lack of a universal transition solution to IPv6, lots of un-optimized solutions are being used for IoT deployment.
- ✓ These makeshift solutions mainly address:
 - IPv6 to IPv4 translation
 - IPv6 tunneling over IPv4
 - Application layer proxies (e.g: data relaying)

Multi-homing

- ✓ A node/network connected to multiple networks for improved reliability.
- ✓ In cases of small IoT LANs, where allotment of address prefixes is not feasible and possible, a proxy based approach is used to manage multiple IP addresses and map them to link local addresses.
- ✓ In another, gateway-based approach is used for assigning link local addresses to the nodes under it.

Source: Teemu Savolainen, Jonne Soininen, and Bilhanan Silverajan, "IPv6 Addressing Strategies for IoT", IEEE Sensors Journal, Vol. 13, No. 10, Oct 2013

- ✓ Providing source addresses, destination addresses and routing information to the multi-homed nodes is the real challenge in multi-homing networks.
- ✓ In case the destination and source addresses originate from the same prefix, routing between gateways can be employed for IoT gateway selection.
- ✓ Presently, IEFT is still trying to standardize this issue.

IPv4 versus IPv6

	IPv4	IPv6
Developed	IETF 1974	IEF 1998
Length (bits)	32	128
No. of Addresses	2^{32}	2^{128}
Notation	Dotted Decimal	Hexadecimal
Dynamic Allocation of addresses	DHCP	SLAAC/ DHCPv6
IPSec	Optional	Compulsory

IPv4 versus IPv6

	IPv4	IPv6
Header Size	Variable	Fixed
Header Checksum	Yes	No
Header Options	Yes	No
Broadcast Addresses	Yes	No
Multicast Address	No	Yes

IPv4 Header Format

1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
Ver			IHL					Type of Service								Total Length															
Identification															Flags			Fragment Offset													
Time to Live								Protocol								Header Checksum															
Source Address (32 bit)																															
Destination Address (32 bit)																															
Options																								Padding							

IPv4

- ✓ The IPv4 emphasizes more on reliable transmission, as is evident by fields such as type of service, total length, id, offset, TTL, checksum fields.

IPv6 Header Format

1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8						
Ver			Traffic Class									Flow Label																									
Payload Length																		Next Header								Hop Limit											
Source Address (128 bit)																																					
Destination Length (128 bit)																																					

IPv6

- ✓ The IPv6 header structure is more simpler as it mainly focuses on the addressing part of the source and destination.
- ✓ It is concerned more with addressing than with reliability of data delivery.

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IoT

- ✓ Internet technology connecting devices, machines and tools to the internet by means of wireless technologies.
- ✓ Over 9 billion 'Things' connected to the Internet, as of now.
- ✓ 'Things' connected to the Internet are projected to cross 20 billion in the near future.
- ✓ Unification of technologies such as low-power embedded systems, cloud computing, big-data, machine learning, and networking.

Origin of Terminology

In the 2000s, we are heading into a new era of ubiquity, where the “users” of the Internet will be counted in billions and where humans may become the minority as generators and receivers of traffic. Instead, most of the traffic will flow between devices and all kinds of “things”, thereby creating a much wider and more complex Internet of Things.

(“The Internet of Things”, ITU Internet Report 2005)

- ✓ The title of the report was “Internet of Things”
- ✓ Discussed the possibility of internet connected M2M connectivity networks, extending to common household devices.
- ✓ Some areas identified as IoT enablers:
 - RFID,
 - Nanotechnology,
 - Sensors,
 - Smart Networks.

Reference: [International Telecommunications Union \(ITU\). \(2005\). *The Internet of Things. Executive Summary* \[Online\]](#)

Alternate Definition

The Internet of Things (IoT) is the network of physical objects that contain embedded technology to communicate and sense or interact with their internal states or the external environment.

Gartner Research

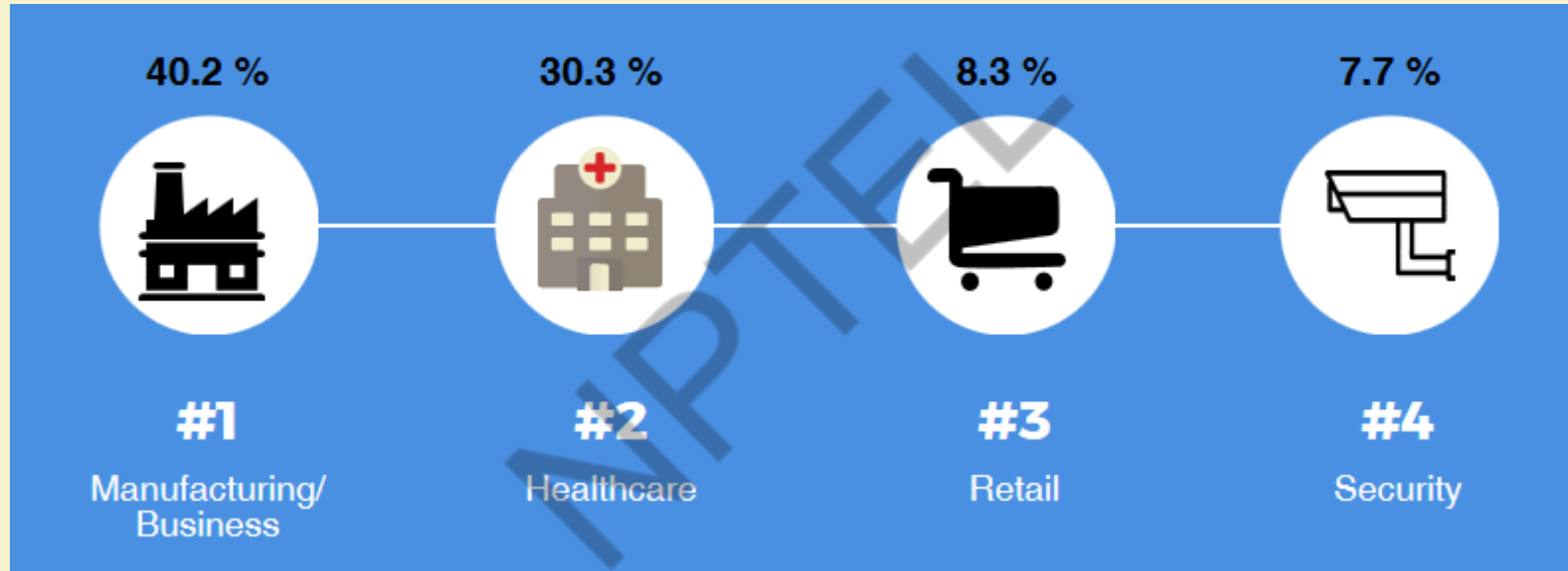
Reference: <http://www.gartner.com/it-glossary/internet-of-things/>

Characteristics

- ✓ Efficient, scalable and associated architecture
- ✓ Unambiguous naming and addressing
- ✓ Abundance of sleeping nodes, mobile and non-IP devices
- ✓ Intermittent connectivity

Reference: Teemu Savolainen, Jonne Soininen, and Bilhanan Silverajan, "IPv6 Addressing Strategies for IoT", IEEE SENSORS JOURNAL, VOL. 13, NO. 10, OCTOBER 2013

IoT Market Share



Source: Intel

✓ Business/Manufacturing

- Real-time analytics of supply chains and equipment, robotic machinery.

✓ Healthcare

- Portable health monitoring, electronic recordkeeping, pharmaceutical safeguards.

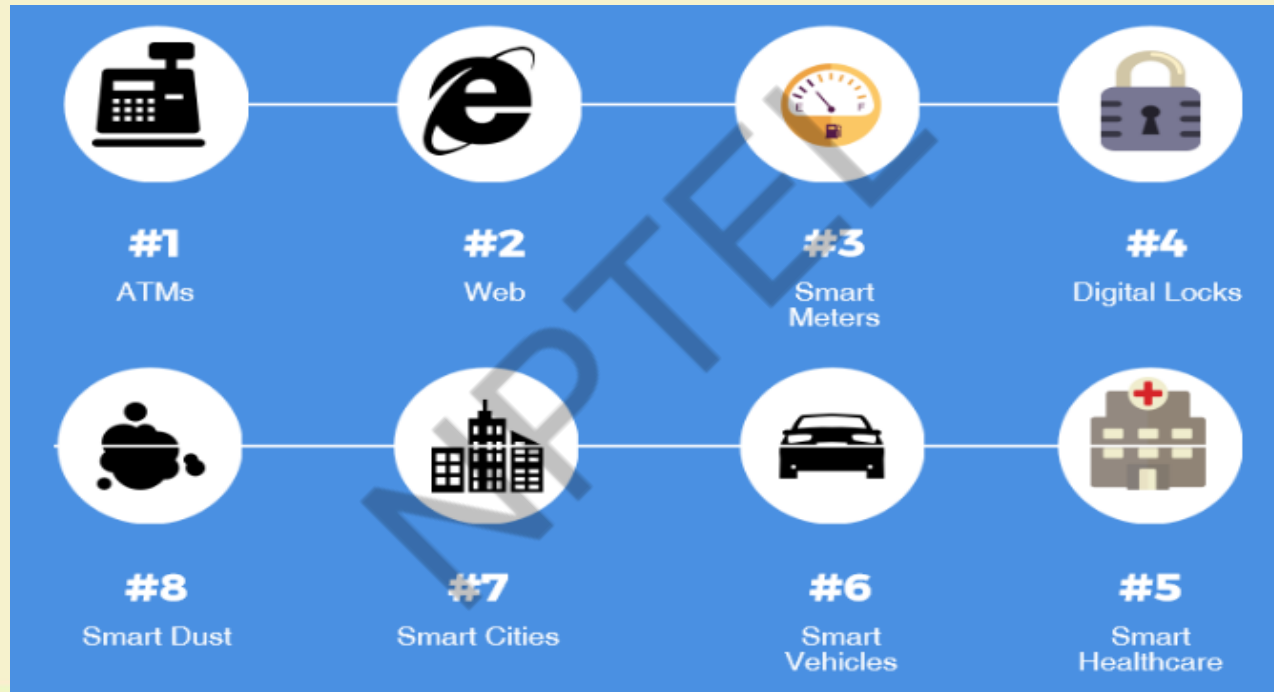
✓ Retail

- Inventory tracking, smartphone purchasing, anonymous analytics of consumer choices.

✓ Security

- Biometric and facial recognition locks, remote sensors.

Evolution of Connected Devices



✓ ATM

- These ubiquitous money dispensers went online for the first time way back in 1974.

✓ WEB

- World Wide Web made its debut in 1991 to revolutionize computing and communications.

✓ SMART METERS

- The first power meters to communicate remotely with the grid were installed in the early 2000s.

✓ DIGITAL LOCKS

- Smartphones can be used to lock and unlock doors remotely, and business owners can change key codes rapidly to grant or restrict access to employees and guests.

✓ SMART HEALTHCARE

- Devices connect to hospitals, doctors and relatives to alert them of medical emergencies and take preventive measures.

✓ SMART VEHICLES

- Vehicles self-diagnose themselves and alert owners about system failures.

✓ SMART CITIES

- City-wide infrastructure communicating amongst themselves for unified and synchronized operations and information dissemination.

✓ SMART DUST

- Computers smaller than a grain of sand can be sprayed or injected almost anywhere to measure chemicals in the soil or to diagnose problems in the human body.

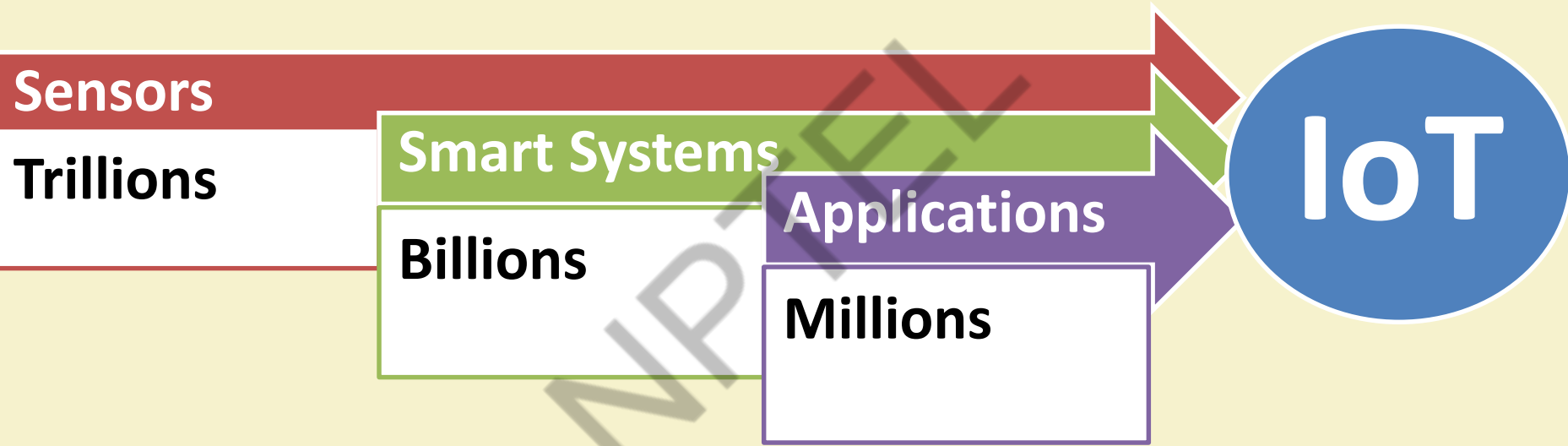
Modern Day IoT Applications

- ✓ Smart Parking
- ✓ Structural health
- ✓ Noise Urban Maps
- ✓ Smartphone Detection
- ✓ Traffic Congestion
- ✓ Smart Lighting
- ✓ Waste Management
- ✓ Smart Roads
- ✓ River Floods
- ✓ Smart Grid
- ✓ Tank level
- ✓ Photovoltaic Installations
- ✓ Water Flow
- ✓ Silos Stock Calculation
- ✓ Perimeter Access Control
- ✓ Liquid Presence

Modern Day IoT Applications

- ✓ Forest Fire Detection
- ✓ Air Pollution
- ✓ Snow Level Monitoring
- ✓ Landslide and Avalanche Prevention
- ✓ Earthquake Early Detection
- ✓ Water Leakages
- ✓ Radiation Levels
- ✓ Explosive and Hazardous Gases
- ✓ Supply Chain Control
- ✓ NFC Payment
- ✓ Intelligent Shopping Applications
- ✓ Smart Product Management

Expected!!



IoT Enablers

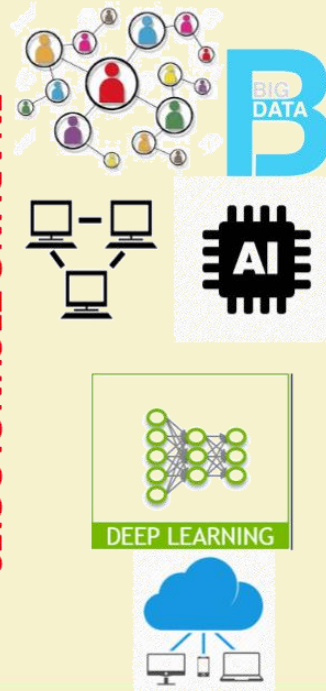


IMPLEMENTATION

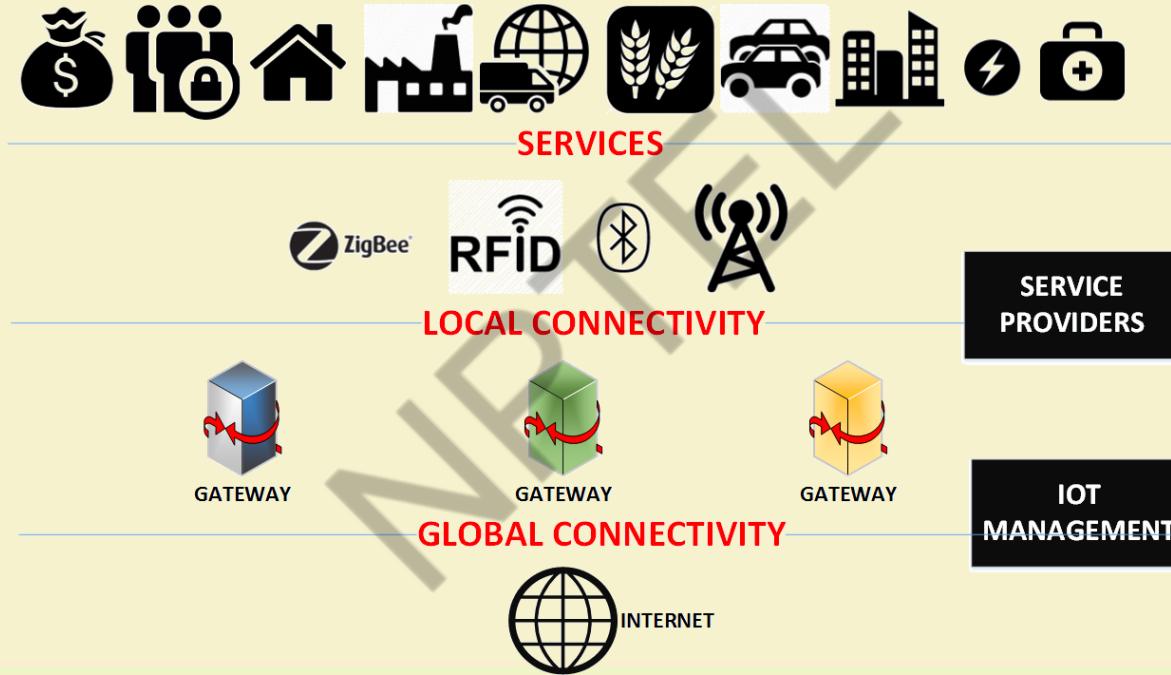


CONNECTIVITY

ENABLING TECHNOLOGIES



Connectivity Layers



Baseline Technologies

- ✓ A number of technologies that are very closely related to IoT include
 - Machine-to-Machine (M2M) communications,
 - Cyber-Physical-Systems (CPS)
 - Web-of-Things (WoT).

IoT vs. M2M

- ✓ M2M refers to communications and interactions between machines and devices.
- ✓ Such interactions can occur via a cloud computing infrastructure (e.g., devices exchanging information through a cloud infrastructure).
- ✓ M2M offers the means for managing devices and devices interaction, while also collecting machine and/or sensor data.
- ✓ M2M is a term introduced by telecommunication services providers and, pays emphasis on machines interactions via one or more telcom/communication networks (e.g., 3G, 4G, 5G, satellite, public networks).

IoT vs. M2M

- ✓ **M2M is part of the IoT, while M2M standards have a prominent place in the IoT standards landscape.**
- ✓ However, IoT has a broader scope than M2M, since it comprises a broader range of interactions, including interactions between devices/things, things and people, things with applications and people with applications.
- ✓ It also enables the composition of workflows comprising all of the above interactions.
- ✓ IoT includes the notion of internet connectivity (which is provided in most of the networks outlined above), but is not necessarily focused on the use of telecom networks.

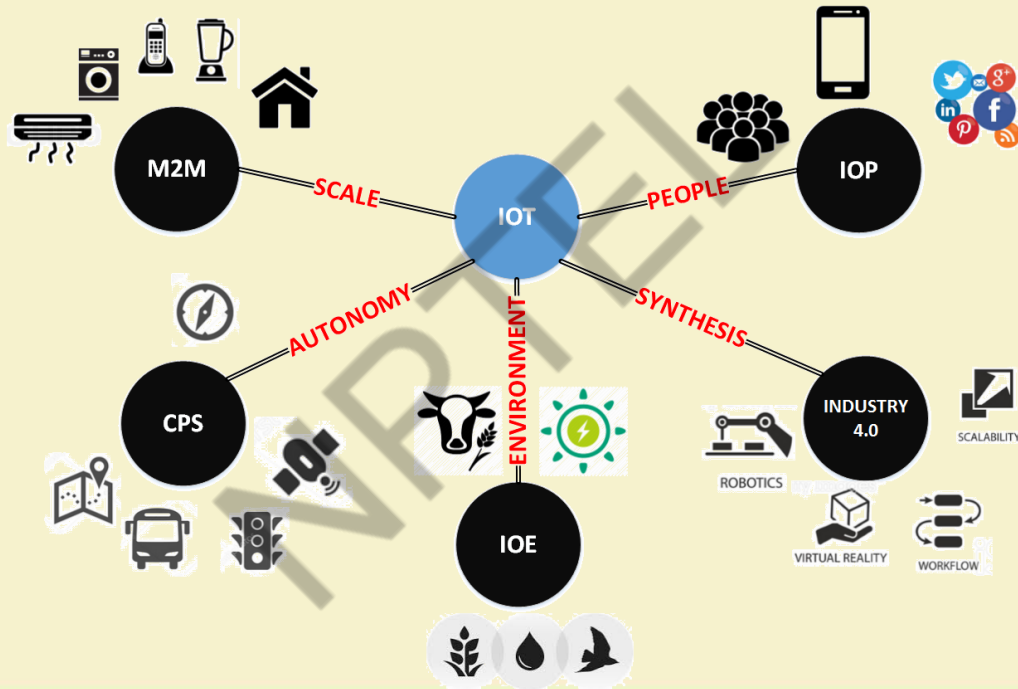
IoT vs. WoT

- ✓ From a developer's perspective, the WoT enables access and control over IoT resources and applications using mainstream web technologies (such as HTML 5.0, JavaScript, Ajax, PHP, Ruby n' Rails etc.).
 - The approach to building WoT is therefore based on RESTful principles and REST APIs, which enable both developers and deployers to benefit from the popularity and maturity of web technologies.
 - Still, building the WoT has various scalability, security etc. challenges, especially as part of a roadmap towards a global WoT.

IoT vs. WoT

- ✓ While IoT is about creating a network of objects, things, people, systems and applications, WoT tries to integrate them to the Web.
- ✓ Technically speaking, WoT can be thought as a flavour/option of an application layer added over the IoT's network layer. However, the scope of IoT applications is broader and includes systems that are not accessible through the web (e.g., conventional WSN and RFID systems).

Terminological Interdependence



Thank You!!

